DACS: An Integrated Digital Agriculture Collaboration and Support System for Poultry Breeder Management, Farmer Education, and Order Processing

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CHAPTER 1 : INTRODUCTION

This chapter sets the context for the study. It discusses the project rationale and context. It summarizes the literature relevant to the research, the research methods and findings, and the problem statement. It also describes the project narrative and details the proposed Digital Agriculture Collaboration and Support System (DACS), its description and objectives, its scope and delimitations, its SWOT analysis and its testing and evaluation framework.

The preliminary investigations of the current state of Dominant Asia Poultry Genetics show the challenges of the current systems and the rationale for integrating the information systems. These sections explain the purpose of DACS in relation to the centralization of the operational records and workflows, the enhancement of inquiries and transaction monitoring, the incorporation of the business rules of the organization, and the provision of services to both the administrative staff and the registered and prospective poultry farmers.

This chapter provides both the importance of the study and the framework within which DACS will contribute to the improvement of information systems, operational effectiveness, service provision, and the quality of decisions in Dominant Asia Poultry Genetics.

A. Project Rationale

Dominant Asia Poultry Genetics is an enterprise from the Philippines that sells Parent Stocks, F1 products, and veterinary supplies. They offer their products to poultry farmers throughout the Philippines. They also have outreach programs for poultry education, and they run and sponsor poultry breeders to obtain their certification. Their expansion has increased the number of customer requests and transactions, and has increased the number of operational records. Dominant Asia Poultry Genetics has implemented the use of technology to modernize their operations. They utilize Google Forms, Google Sheets, and Facebook to aid in processing customer requests, capturing and registering payments, and certifying poultry breeders. Unfortunately, since each of these systems are stand alone, operational records are incomplete, and staff are required to consolidate and authenticate the records captured in numerous systems.

Interviews and analysis of operations revealed that there are multiple problems that customers and the enterprise itself are contending with. These include incomplete fragmentation of records, multiple records for the same customer, duplicating the entry of customer information, manual checks to confirm completion of seminars, eligibility to certify breeders, inefficient tracking of customer orders and inquiries; and lack of a single platform to access services offered by the enterprise. The cumulative effect of these problems is increasing the administrative burden, obstructing business operations, and decreasing the quality of service offered.

The proponents present the Digital Agriculture Collaboration and Support System (DACS) to provide a solution to these problems. It is a web-based and mobile adaptable system that consolidates core operational processes. The research adopts a Minimum Viable Product (MVP) approach with operational modules that include: User Management, Customer Information Management, Order Management and Payment Recording, Seminar Management, and the Certified Breeder Registry. The preliminary investigation identified the above modules to be critical in addressing the core operational problems.

DACS incorporates Inquiry Management, Historical Data Management, and Analytics and Reporting. These modules offer the organization several functions that include customer service, operational management, and decision making. The Inquiry Management module provides farmers the capability to submit inquiry tickets to DACS, as well as monitor the status of their inquiry ticket. Administrators use DACS to categorize, assign, and update the status of tickets. The module also provides a channel to post ticket resolutions and responses.

The research adopts a phased development approach to the new system. It offers Dominant Asia Poultry Genetics an integrated information system that enhances operational information and management, supports organizational business rules, and improves the monitoring of inquiries, thus providing a better experience for the proposed system users.

a. Research Context

This research falls under Agricultural Information Systems and the use of IT in the improvement of agricultural systems and services. The digital transformation of business processes and services has led to the proliferation of information systems. These systems facilitate the management of operational data and workflows, enhance the quality of decisions, and improve the overall service.

The Philippines' Dominant Asia Poultry Genetics provides Parent Stocks, F1 products, veterinary products, and services to poultry farmers. This includes poultry education seminars and breeder certification. In support of its business, the company employs various digital offerings such as Google Forms and Google Sheets, and other communication products such as Facebook and email. While these existing technologies support particular business processes, they do so in isolation as no integrated operational platform is available.

With the continued growth of the business, the limitations of the current operational setup has become more evident. Information regarding customers, seminar participation, breeder certification, customer queries, sales orders, and payments exist in multiple disconnected platforms. Information consolidation and verification is the responsibility of administrative staff, which has led to the poor integration of information, redundant records and data, manual operational verification, limited status visibility of inquiries and order fulfillment, prolonged delays, and suboptimal customer service.

In this case, the proposed Digital Agriculture Collaboration and Support System (DACS) acts as a centralized, integrated, web-based, and mobile information system that captures core business processes within the organization. This system also encompasses the management of Customer Information, the Management and Recording of Orders and Payments, Management of Seminars, the Certified Breeder Registry, Management of Inquiries, Management of Historical Data, and Analytics and Reporting. Inquiries will be lodged and tracked within DACS, while responses and resolutions will be provided through the official email of the organization.

The proposed system, by aiming to enhance operational effectiveness and overall service experience for the registered and potential poultry farmers, as well as for Dominant Asia Poultry Genetics, and by aiming to reinforce and enhance the overall experience, is designed to improve the standardization of operational processes and workflows, enable systematic management of business rules of the organization, and improve management of inquiries.

i. Review of Related Literature

This section outlines studies and concepts that aid the construction of the Digital Agriculture Collaboration and Support System (DACS). The literature examines the application of digital technology in the field of agriculture, with specific regard to poultry, along with information systems to enhance data management, streamline processes within the organization, strengthen stakeholder relationships, maintain certification logs, and aid rationalization in operational activities and decisions.

It includes literature analyzing widely known challenges faced by agricultural enterprises, such as disjointed information systems, manual processes and record keeping, limited information accessibility, and an emerging demand for comprehensive digital solutions. These studies assist the authors in establishing the practical and theoretical basis for DACS and the rationale for a unified system to manage customer records, poultry breeders, educators, inquiries, orders, payments, data, reports, and operational history.

i.a Digital Transformation in Agricultural and Poultry Operations

The agricultural sector has been positively impacted in several areas due to the changes in digital technology. As agricultural companies grow, traditional methods for keeping records, managing transactions, and coordinating activities with stakeholders will not be able to keep up. Digital transformation helps by unifying business systems and allowing faster, more accurate information and transactions to be processed for better, more informed decisions.

In companies that deal with poultry, information management plays a huge role in managing poultry stock, customer service, operations, and the overall success of the business. According to Akinde et al. (2025), there are many issues faced by poultry companies due to the use of manual record keeping, such as having to wait for information to be shared, reports to be created, and the data to be accurate. These issues cause a decline in how well the company operates and in the management decisions that are made. The authors of the study called for the development of integrated digital systems that can handle the fragmented digital systems of the poultry sector.

The same issues are faced by poultry companies that use non-standardized poultry systems, as noted by Geetha et al. (2020). Having a non-standardized system will lead to wasted time and duplicated data. The authors noted that digital systems can unify organizational data and create standards.

The use of technology in poultry farming to manage operations is growing due to the digital transformation of the industry. Kartanova et al. (2025) provided examples of poultry management systems that have a decision support system and a centralized data system. These systems are able to improve management by structuring system operations and providing data to managers in order to help them fulfill their monitoring and business operational needs.

The studies show that digital transformation means more than simply automating stand-alone activities. Transforming digitally means the integration of workflows, functions, and records at the organizational level and within a shared digital space. The literature relates to the introduction of centralized, integrated systems that enhance the level of efficiency, accessibility, accuracy of records, and performance at the organizational level. The findings offer a basis for the creation of DACS as an integrated system to respond to the problem of fragmented workflows and the management of organizational information at Dominant Asia Poultry Genetics.

i.b Data Management and Centralized Information Systems

Data management is vital to the majority of most data reliant information systems. As an organization grows, the task of maintaining accurate, complete, and timely records becomes more difficult. Poor data management leads to inadequate records that may be inconsistent, duplicated, incomplete, or difficult to find. Poor data management also negatively impacts the efficiency of systems and operations, and the monitoring and evaluation of systems and operations is severely restricted. Most systems and operations will suffer from a lack of data driven decisions.

Zheng et al. (2021) described the use of information systems in the poultry industry and the solutions offered by the cloud to fragmented databases. With cloud information systems, fragmented operational data is stored in one place and becomes more consistent and reliable. The cloud also allows users to access, share, and update information.

Gunawan and Kuswanto (2024) showed that poor data management results in unreliable records, and reports and information management is key to improving record accuracy and the efficiency of operations and management.

Data from poultry operations is becoming larger and more complex and demonstrates the need for advanced data processing and analytics. Bumanis et al. (2022) described how the poultry industry collects a lot of operational data, but is typically unable to organize, process, and analyze the data to generate actionable information. Centralized information systems overcome this challenge.

Hongqian et al. (2016) elaborated on how cloud-based data-management systems maintain long-term storage of operational data. Their research revealed how centralized digital systems store historical data and offer avenues for authorized personnel to access and amend records using web- and mobile-based systems. This positively impacts data continuity and availability and enhances organizational memory and the ability to engage in diverse operational processes.

The studies included in the review articulate how centralized information systems enhance data quality, availability, and continuity and help improve productivity within an enterprise. The cloud-based systems integrate, control, analyze, and make data available for use to facilitate everyday processes and managerial functions. Within this context, the fragmentation of data and duplicate systems, along with the absence of historical data concerning the operations and performance of the organization, justify the need for the development of DACS as a centralized system.

i.c Process Automation and Operational Efficiency

Operational efficiency is an important goal for many organizational information systems. This is particularly true for those systems that manage large volumes of interconnected records, customer transactions, and interact with various stakeholders. The boundaries of manual processes become clearer as the organization grows. For instance, the manual workflows consume more time and more administrative efforts. Where there is delay, repetition, and inconsistency, there is a negative effect on productivity.

According to Akinde et al. (2025), most of the manual record keeping in the poultry sector leads to slow processing of data with poor coordination of the various functions of the business. The authors argued that an integrated information system offers a centralized location to eliminate unnecessary duplication and improve the efficiency of workflows. The authors suggest that the barriers to productivity can be reduced through improved workflow and better coordination of the operational data.

Geetha et al. (2020) confirmed that lengthy processing, duplication of data, and operational errors are a direct result of relying on manual records and fragmented methods. Their study mentioned that, in addition to the reliability and consistency of business operations, integrated enterprise systems increase productivity by standardizing and interlinking various functions of the organization.

Kagona et al. (2021) studied the impact of communication challenges and the sluggish processing of information in poultry operations that rely on reporting done manually. Their findings indicated that information systems have the potential to improve the access of stakeholders to up-to-date operational data, automate routine tasks, and enhance stakeholder communication. These systems help organizations better manage resources, expedite service, and elevate the operational process.

The literature shows process automation positively impacts time-consuming administrative tasks that are repetitive and follow a set of rules. Automation lowers the frequency of manual tasks, lowers the risk of human error, and allows staff to concentrate on processes that require more advanced skills, such as critical thinking and executive functions. However, automation does not mean the absence of manual tasks. On the contrary, it provides assistance to employees by automating workflows, applying business logic, and enhancing the precision and reliability of operational data.

These results are the foundation for developing DACS as a system with the capability to provide automation for select organizational tasks, such as seminar prerequisite verification, breeder eligibility, order status, payment reporting, ticket inquiries, and operational reports. By increasing the automation of manual tasks and enhancing the inter-related modules of DACS, it is anticipated that Dominant Asia Poultry Genetics will experience increased operational efficiency and improved service.

i.d Digital Farmer Education and Stakeholder Engagement

The ongoing evolution of communication technologies and agricultural expertise has a positive impact on productivity, the ability to make informed decisions, and the acceptance of best practices by the agricultural community. With greater diffusion of digital technologies and services, information systems are being utilized to not only manage day-to-day activities but also offer educational services, facilitate stakeholder participation, and extend services to the end user.

According to Awotunde et al. (2025), the increasing integration of mobile technology for the purpose of information dissemination to poultry farmers has been noted. Timely information through digital communication to the poultry farmers has enhanced their capacity to make informed decisions regarding the management of their poultry. Better access to information in the farming community has lessened the information delivery lags and has improved the dissemination of information and has the potential to improve the management practices of farms.

Digital learning platforms as per Saengwong and Koksantia (2020), can provide poultry farmers with learning materials that are well structured and simpler to learn. Several mobile applications may include information pertaining to the management of poultry, the prevention of diseases in poultry, the poultry feeding regimes and other relevant topics in agriculture. This study revealed that the simplified digital learning environments offer farmers greater access to knowledge and enhance the usability of agricultural educational training services in a given community.

Modern information systems must also consider the engagement of their designed systems. Organizations need to have well structured systems through which their users can pose inquiries, offer concerns and provide feedback. Compared to the fragmented and informal communication systems, a centralized digital system can provide a more structured system for stakeholders to record their interactions and allow their concerns to be addressed.

Merging educational and stakeholder-support functionalities in one platform can improve how information is disseminated and accessed. For instance, in DACS, farmers can access seminar materials, track their progress, view the FAQ Knowledge Base, and submit questions, all from the same account. DACS logs and tracks the progress of submitted tickets, while responses and resolutions are communicated through the organization’s email.

The analyzed literature reveals the significance of digital farmer education and of systematically building stakeholder engagement in agricultural information systems. Systems that incorporate learning and engagement support mechanisms can respond to the educational concerns of farmers, enhance their involvement, and facilitate better communication of the agricultural organization with the stakeholders. The recommendations to include Seminar Management, the FAQ Knowledge Base, and Inquiry Management in DACS are justified.

i.e Digital Certification, Traceability, and Compliance Management

Certification and compliance management are crucial to agricultural firms because they provide accountability, transparency, and trust with stakeholders. As firms grow, keeping track of certification records and compliance requirements becomes more difficult and more error-prone. Digital systems offer a more systematic way to manage certification records, requirements, and documents that need to be maintained.

Zülch et al. (2024) highlighted the role of digital systems in supply chain management and agricultural certification record traceability. Their research showed that, with digital records, stakeholders had much easier access to certification records and certification documents and, therefore, greater confidence and trust in the certification process.

Combined traceability and compliance management systems tackle the difficulties of incomplete records and the need for constant manual verifications. Centralized systems allow firms to record the history of their certifications, track the status of their certifications, retain all necessary documentation, and provide information for verification or audit requests. These systems are valuable if certification is a requirement for a firm to offer programs or services to the public.

Digital management of certification records enhances the management of certification programs because, less time is spent reviewing records, greater clarity and confidence is achieved in compliance, and greater program efficiency is attained. Compliance with eligibility requirements, expiration, and/or validity periods may be managed with automated systems. Such systems, along with the adherence of a system to defined business rules, may also increase the trust placed in certification decisions.

The literature shows that digital systems for certification and traceability increase accuracy of records, ease verification, and enhance compliance tracking. These results justify the creation of the Certified Breeder Registry in DACS. The purpose of the registry is to consolidate all breeder records, track eligibility and certification status, track validity periods, and facilitate the enforcement of the organization's certification rules, all through an integrated system.

i.f Analytics, Decision Support, and Mobile Accessibility

Digital data has the potential to enhance performance reviews, operational assessments, and strategic evaluations. Information systems of today can do much more than record data. They can change data into operational analytics, reports, and diagrams that can be used to make informed operational decisions. Organizations can use these systems to find patterns, analyze performance metrics, and address operational issues more effectively.

According to Leovin et al. (2026), the use of descriptive analytics in poultry information systems has grown significantly. Their research indicated that the use of dashboards and other analytics tools helps organizations to structure their operational data and displays it in a format that enhances management decision-making. Furthermore, analytics systems that represent performance metrics in a visual format enhance the organizations' focus of the management planning, monitoring, and operational activities.

Bumanis et al. (2022) show that the lack of properly structured, processed, and interpreted data within poultry organizations has been recognized as one of the major issues facing these organizations. Information systems that can be centralized and contain embedded analytics have the potential to change operational data into insights that can enhance the evaluation of performance, control of operations, and the management decision process.

According to Handojo et al. (2025), dashboard systems have the added benefit of being used to monitor operational activities. Their research demonstrates that dashboards are systems that present information in a clear and easy to understand format for the user to identify trends and analyze metrics, and may help address performance issues. Compared to traditional tabular formats, graphs and other visual diagrams enhance operational monitoring.

Mobile accessibility must be prioritized within agricultural information systems. Awotunde et al. (2025) explained that mobile tools expanded digital resources and services for farmers without geographical constraints. Farmers can now more conveniently access organizational information, complete transactions, check the status of services, and engage with service providers via mobile apps.

Mambile et al. (2018) said that the adoption of a web-based agricultural platform depends on the platform's usability and the user's experience. Their research asserted that web-based platforms, which are mobile and easily designed to meet the users' needs, are more readily adopted. This is even more pronounced for agricultural users who mainly access the internet services via their mobile.

The literature indicates that analytics and decision support systems, in conjunction with mobile accessibility, represents key components of contemporary agricultural information systems. Operational visibility and the facilitation of decision-making are achieved through the application of dashboards and data visualizations complemented by user participatory designs that are user friendly and mobile accessible. This information substantiates the creation of a mobile-friendly DACS, which enables access to organizational services, operational dashboards, and dashboards.

The literature promotes the need for mobile accessibility, analytics, and other components of a digital framework such as the digital transformation of services, the centralization of data, automation of processes, and digital services for farmers and certification personnel. To the best of the author's knowledge, none of the literature offered a comprehensive platform that integrated the management of customer data, prerequisite-based seminar education, the management of breeder certification, and the monitoring of inquiry tickets, as well as the recording of orders and payments, the management of legacy data, and operational analytics in a poultry genetics corporate environment, which justifies the construction of the Digital Agriculture Collaboration and Support System (DACS).

ii. Research Methods and Findings

The proponents built a general picture of the business environment for Dominant Asia Poultry Genetics and conducted initial data gathering to aid in their evaluation. These steps were meetings with stakeholders to discuss the processes and semi-structured interviews with the business owner and admin staff, and some poultry farmers via Google Meet. The proponents aimed to learn existing processes and problems, unmet user needs, system gaps, and how the organization's present processes would benefit from an information system. They conducted semi-structured interviews, and relied on the results for their thematic analysis. Results were used to interpret the problems and needs that were used to formulate the proposed system.

The findings reported many unintegrated digital platforms used to run the organization's operations. Google Forms, Google Sheets, email, Facebook Groups, and Facebook Messenger are used to store customer details, seminar participation, breeder certification, orders, payments, inquiries, and business communications. Therefore, the Administrative Staff are required to repeatedly validate the accuracy and completeness of the records on unintegrated platforms.

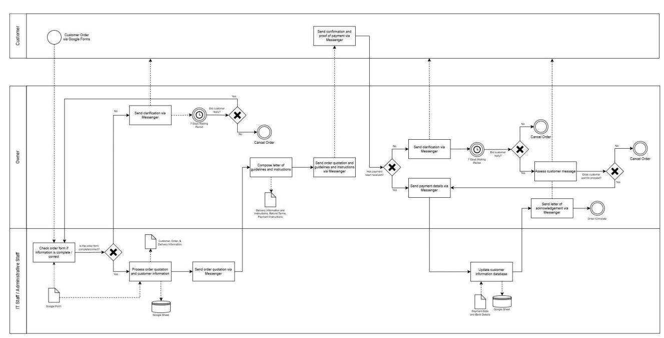

Fig. 1.1 Order Handling of F1 Chicks Process Flow

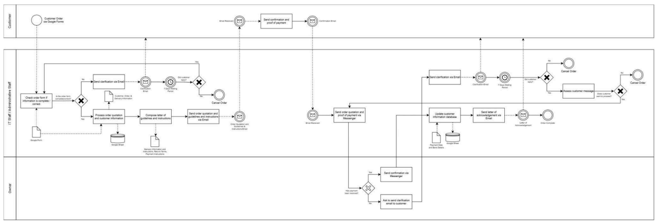

Fig. 1.2 Order Handling of PS Process Flow

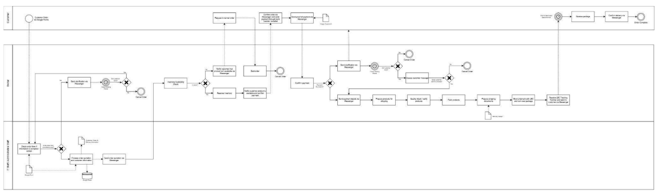

Fig. 1.3 Order Handling of Veterinary Products Process Flow

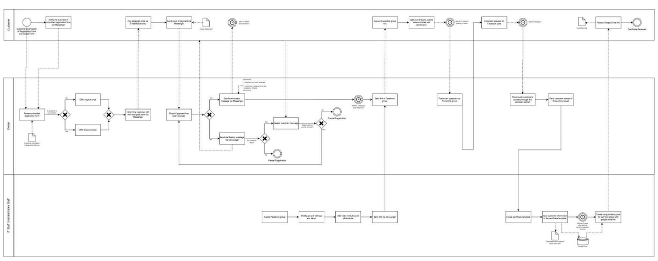

Fig. 1.4 Dominant Asia Seminar Certification Process Flow

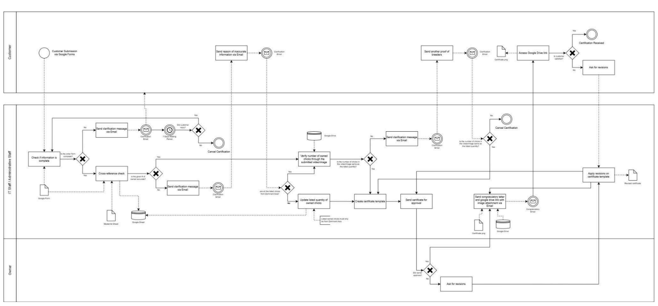

Fig. 1.5 Dominant Asia Breeder Certification Process Flow

The fragmented nature of the existing process was described by several participants. One participant said

"The product information was posted on Facebook, the order was submitted through Google Forms, and the confirmation was sent through Messenger. The whole process felt scattered."

Records Management was another significant issue cited by the participants. This included duplicate, inconsistent records. Several participants cited having to provide the same information multiple times to register for seminars, submit inquiries and orders. One participant said,

"I already submitted my personal and farm information in the seminar form, but I had to enter the same details again in the inquiry form and order form."

The lack of a centralized system impedes the organization's ability to track customers and make operational and strategic decisions.

Further inefficiencies were discovered in handling customer orders and inquiries. Customer inquiries arrive mainly via Facebook Messenger. Customer orders, Seminar records. Payment records, and eligibility Information are all housed in separate systems. Messenger lacks a formal way to assign inquiries a reference number or categorize, track, or monitor the status of the inquiries. Stakeholders mentioned that during peak periods, the backlog of inquiries may exceed one hundred (100) and be left unread, which increases the chance of missed customer inquiries and transactions.

The research also showed that seminar attendance records and seminar order records are kept in separate places. This means that the Administrative Staff, who orders Parent Stocks, must first confirm that the seminar modules have been completed by the farmer who is placing the order. One respondent said,

"When I tried to order Parent Stocks, the staff could not immediately confirm my eligibility because my seminar records were stored separately from my order details."

Sector fragmentation results in increased tasks for the staff to manually check and confirm records. This is reported to have resulted in poor and inconsistent service to the customers. Multiple versions of the same form in earlier messages caused confusion and poor service from the Administrative Staff who were asked to confirm which is the correct form after checking multiple records. Several participants reported difficulty identifying which is the correct form.

Concerns regarding accessibility and user experience were also raised. The organization primarily caters to poultry farmers from different regions of the Philippines. Many farmers utilize digital services via mobile devices and possess different levels of digital literacy. Farmers do not have a centralized platform to track their seminar attendance, Breeder eligibility, Inquiry ticket status, purchase orders, verify payments, and view transaction histories. As a result, farmers are compelled to message, email, or call the organization to receive status updates. This increases the Administrative Staff's workload. One of the participants summarized:

“It would be easier to use one account where I can register for seminars, check my eligibility, submit inquiries, order products, and track my transactions.”

Based on the results, the organization does not have control over information. The organizational processes are dependent on repetitive data entry and manual verification. Integrations with other systems are lacking, and the user interactions are managed by multiple independent systems. This leads to a larger administrative burden, decreased efficiency, and an inability for the organization to cope with an increasing user demand and deliver adequate services to the poultry farmers across the nation within a reasonable timeframe.

The results were the foundation for the Digital Agriculture Collaboration and Support System (DACS) design. The intended purpose for the proposed System was to support an integrated platform for various Customer Information, Order Management and Payment Recording, Seminar Management, the Certified Breeder Registry, Inquiry Management, Historical Data, and Analytics and Reporting systems. Farmers, via the DACS platform, will be able to submit and manage inquiry tickets and view responses and resolutions via the organizational email system.

Centralization of DACS’s operational processes aims to achieve the following key objectives: consistent data; less repeated data entry and manual cross-referencing; stronger enforcement of organizational business rules; better monitoring of inquiry status; and improved operational efficiency of Dominant Asia Poultry Genetics.

a. Statement of the Problem

General Problem

Dominant Asia Poultry Genetics' business operations involve several disparate digital systems for customer relationship management, seminar management, breeder certification, order management, payment management, customer support, and business communication. Without a system of integrated information, Dominant Asia Poultry Genetics faces incomplete records, repeated data entry and verification, sub-optimal business operations, and poor customer and operations information visibility. This system fragmentation negatively impacts the administrative staff of Dominant Asia Poultry Genetics and the poultry farmers, both registered and potential, who use its services.

The focus of this project is the design and development of the Digital Agriculture Collaboration and Support System (DACS). This system will be a fully centralized, web-based digital system, accessible via mobile, that will increase the efficiency of business operations, and management of information, and improve the monitoring of customer inquiries and the overall accessibility of services offered by Dominant Asia Poultry Genetics for registered and potential poultry farmers.

Specific Problems

1. How can DACS reduce information fragmentation, duplicate records, data entry, and manual cross-referencing by integrating customer, order, payment, seminar, breeder certification, inquiry ticket, and historical data?
2. How can the proposed system facilitate better verification of customer orders, payment, participation in seminars, breeder certification, certification validity, inquiry tickets, and verification of other operational activities of Dominant Asia Poultry Genetics?
3. How can the proposed system enhance visibility and aid management through dashboards and operational reports?

4. How can the proposed system automate and codify the organization's business rules, specifically the seminar prerequisites for ordering Parent Stock and the eligibility and certification validity of breeders?
5. How can the proposed system offer registered and potential clients a mobile and user-friendly way to access the organization's services to order and track progress, participate in seminars, check eligibility and certification status of breeders, and view their transaction history?
6. How can the proposed system offer a way to submit inquiry tickets, track ticket status, and access a Frequently Asked Questions Knowledge Base to reduce the need for multiple communication methods? Responses to inquiries and ticket resolutions will be provided via email.
7. Would the proposed system noticeably improve the user experience of registered and potential poultry farmers after twenty-one (21) days of continuous use compared to the experience prior to using the system?
8. What is the user acceptance level and perceived software quality of the proposed DACS?

   8.1 Functional Suitability

   8.2 Performance Efficiency

   8.3 Reliability

   8.4 Usability

B. Project Narrative

The preliminary investigation conducted at Dominant Asia Poultry Genetics revealed key operational issues that need to be solved. The Digital Agriculture Collaboration and Support System (DACS) focuses on the design and development of a flexible, integrated, easy-to-use, web-based and mobile friendly system that centralizes core operational processes. These processes include User Management and the Management of Customer Information, Orders and Payments, Seminars, the Certified Breeder Registry, Inquiries, Historical Data, as well as Analytics and Reporting. The DACS aims to develop an integrated, flexible and user-friendly system that effectively and efficiently manages core business processes and improves the quality of services offered both to the employees and customers of Dominant Asia Poultry Genetics. It replaces fragmented information management with a centralized database.

The system is designed for two main user groups. For the personnel of Dominant Asia Poultry Genetics, DACS offers a centralized system for managing customer records, tracking seminar participation, assessing breeders' eligibility and certification, processing customer orders, capturing and validating payments, tracking inquiry tickets, maintaining the FAQ Knowledge Base, and generating operational reports. DACS will capture, classify and track ticket statuses. Communication of responses and resolutions to inquiries will be done via the email channel designated by the organization.

The system provides registered and potential poultry farmers a mobile-friendly platform to manage customer profiles, access seminar modules, track progress, order products, upload payments, check breeder status and certifications, submit tickets, view the FAQ Knowledge Base, and track orders, payments, inquiries, and other transaction records in one account.

To design a system proposed to be constructed within the timeline of the capstone project, the Minimum Viable Product (MVP) Model was adopted. The core operational modules of User Management, Customer Information Management, Order Management and Payment Recording, Seminar Management, and the Certified Breeder Registry, have been included in the MVP. These modules address operational issues directly. The other operational modules of Inquiry Management, Historical Data Management, and Analysis and Reporting, complement the core modules by structured recording of inquiries, centralized historical records, and operational reports, respectively.

The proposed DACS (Digital Agricultural Collaboration and Support System) seeks to integrate operational processes in one information system. This will reduce information silos, eliminate manual verification and repeated input of the same data, improve consistency with organizational business processes, strengthen monitoring of inquiries and transactions, and provide an improved and streamlined service for employees of Dominant Asia Poultry Genetics and registered and aspiring poultry farmers.

a. Description and Objectives

The Digital Agriculture Collaboration and Support System (DACS) uses web-based and mobile-friendly platforms to combine and streamline the processes of Dominant Asia Poultry Genetics. DACS integrates User Management, Customer Information Management, Order Management, and Payment Recording, as well as Seminar Management, the Certified Breeder Registry, and Inquiry Management. It also includes Historical Data Management, Analytics, and Reporting. Combining these operational processes enables the system to unify and streamline the organization's workflow. The combined streamline processes, coupled with operational efficiency and service improvements, alleviate the fragmentation of information, repeat manual data verification, and data entry.

The system is designed to support two primary user groups. The first user group is the personnel of Dominant Asia Poultry Genetics. The system, for this user group, provides tools for the centralized management of customer records and participation in seminars; the verification of breeders and their certification; the processing and certification of payment; the monitoring of inquiry tickets and management of the FAQs Knowledge Base; and the generation of operational reports. Tickets are monitored on DACS, while inquiry responses and resolutions are conveyed via the organization's official emailing channel.

The second user group of the system is the registered and prospective poultry farmers. For this user group the system provides a mobile-friendly interface to manage customer profiles; access seminar modules; track progress; place product orders; upload proof of payment; check eligibility and certification; submit inquiry tickets; access the FAQs Knowledge Base; and monitor orders, payment, inquiries, and other transaction records. The features listed for this user group are accessible through a single account.

To comply with the project's time constraints, the proponents used the Minimum Viable Product (MVP) which involves the implementation of core operational modules with MVP including User Management, Customer Information Management, Order Management and Payment Recording, Seminar Management, and the Certified Breeder Registry. Other features such as Inquiry Management, Historical Data Management, and Analytics and Reporting aid the customer support process by improving the monitoring of data and operations, and assist management and operational evaluation and reporting.

The primary aim of the study is to create and establish the Digital Agriculture Collaboration and Support System (DACS). This system is a web-based and mobile-accessible integrated system developed to enhance the management of information and operations, the monitoring of inquiries and the delivery of services to staff of Dominant Asia Poultry Genetics as well as registered and potential poultry farmers.

Specifically, this study aims to:

a. To consolidate customer, order, payment, seminar, breeder certification, inquiry ticket, and historical data into one information system to alleviate data fragmentation, record duplication, repetitive data entry, and manual cross-reference checks.
b. To enable Administrative Staff with systematic workflows for processing customer data, verifying payments and orders, tracking seminar statuses, and managing the certifications and eligibility of breeders, as well as the statuses of inquiry tickets.
c. To utilize automation for certain organizational business rules, specifically the completion of Seminar Modules 1, 2, and 3 prior to the ordering of Parent Stock, and the monitoring of breeder certification and eligibility.
d. To enhance information accessibility and aid management with decision support through the availability of role-based dashboards, searchable records, and operational reports.
e. To offer registered and prospective poultry

farmers a straightforward mobile application to access seminar modules, track seminar progress, place orders, payment order uploads, and check Breeder certification and eligibility status, as well as view transaction records.

f. To offer an organized Inquiry Management module to farmers that allows for the submission of inquiry tickets, access to the FAQ Knowledge Base, ticket status monitoring through DACS, and email communication of inquiry ticket response and resolution from the organization.
g. To substantiate the extent of user experience reported before and after the implementation of DACS using a twenty-one (21) day DACS usage period.
h. To analyze the proposed Digital Agriculture Collaboration and Support System (DACS) using a Comparative Analysis, Functional Testing, Integration Testing, User Acceptance Testing (UAT), and the ISO/IEC 25010 Software Quality Evaluation from the perspectives of:
    i. Functional Suitability
    ii. Performance Efficiency
    iii. Reliability
    iv. Usability

b. Scope and Limitations

This research assesses the design, development, and evaluation of the Digital Agriculture Collaboration and Support System (DACS) invested by Dominant Asia Poultry Genetics. The system seeks to integrate multiple operational activities into a core system. These activities include User Management, Customer Information Management, Order Management, Payment Recording, Seminar Management, Certified Breeder Registry, Inquiry Management, Historical Data Management, and Analytics and Reporting. The system will be developed in a web-based, mobile-optimized, and user-friendly format.

To ensure the practicality of the system and the time restrictions of the capstone project, the Minimum Viable Product (MVP) was implemented. The core operational requirements explained in the preliminary investigation focused on the development of the MVP. These requirements consisted of User Management, Customer Information Management, Order Management and Payment Recording, Seminar Management, and the Certified Breeder Registry. These components and modules fulfilled the core operational requirements of the system and established the core operational activities of the Institute. The activities included the system's management and processing of customer information, customer orders, the capture and unblocking of payment orders, the management of webinar participation, and the verification and certification of breeders. The modules also included the automation of the Institute's business rules and operations.

Supporting functionalities like Inquiry Management, Historical Data Management, Analytics, and Reporting build on top of the core modules. Inquiry Management allows farmers to create inquiries, access the FAQs Knowledge Base, and track the status of their inquiries in DACS. Authorized Administrative Staff can classify, assign, track, and edit the status of the inquiries. Responses and resolutions will be routed via the organization's email. Historical Data Management consolidates various forms of records, including Customers, Orders, Payments, Seminars, and Breeder Certifications. Analytics and Reporting contain dashboards, and operational reports, for monitoring the organization and providing support for management decisions.

The system will be assessed using Comparative Analysis, Functional Testing, Integration Testing, User Acceptance Testing, and ISO/IEC 25010 Software Quality Evaluation. Assessment participants will be the Owner/Executive, specific Administrative Staff, and IT staff, along with 30 registered and potential poultry farmers.

The operational processes of Dominant Asia Poultry Genetics will bound this research. The proposed system will not have any Enterprise Resource Planning functionalities for accounting, payroll, purchasing, inventory, logistics and supply chain functions. Also, integrated online payment systems for real-time electronic payment processing will not be included, along with offline capabilities, third party Learning Management System, and implementation for other organizations besides Dominant Asia Poultry Genetics. Payment transactions will be processed through the organization's existing financial processes. DACS will support the submission of proof of payment, record payments, and facilitate manual payment verification by authorized staff.

c. SWOT Analysis

Strengths	Includes a versatile range of functionalities, such as the management of customer information, payments, and orders, certified breeder registration, seminar management, inquiries, and historical data, as well as analytics and reporting, all on one platform.
	Completes data unification at the organizational level and eliminates the need for cross-referencing information, as well as manual redundancy, and duplicate records.
	Enables the automation of seminar prerequisite validation, breeder eligibility, and certification checks.
	Structures the submission of inquiries via tickets, allows the submission and monitoring of tickets, and provides an FAQ Knowledge Base. Responses and answers are sent via email.
	Provides operational reports and dashboards that aid the monitoring and provides data for management-related decisions.
	Enables secure data management through the implementation of role-based access control.
Weaknesses	Requires a reliable and stable internet connection to access the system and conduct operations.
	The first migration of historical data may be a time-intensive requirement due to data validation, cleansing, and consolidation processes.
	Users with low digital skills may need to attend a training and orientation session before using the system.
	System dependency and reliability will be a requirement for ongoing support and maintenance.
	The system's reliance on ticket submissions and the monitoring of ticket status limits the organization's ability to provide quick responses and answers to inquiries.
Opportunities	Modernizes the agricultural sector and digitally transforms the organizations within the sector.
	Provides potential for further enhancements, such as a dedicated mobile application, SMS notifications, and integrated online payment systems.
	Promotes the use of digital services for seminar participation, order placement, management of inquiries, and breeder certification among farmers.
	Supports the design and implementation of organizational plans through the provision of actionable data.
	May provide an example for digital

	transformation for other agriculture and poultry businesses.
Threats	Lack of acceptance for automation from staff and other users who prefer manual tasks.
	Cybersecurity and data privacy challenges that may impact system security.
	Remote regions may create limited system accessibility due to lack of internet.
	Changes in organization may require adjusting to a flexible system.
	New agricultural management systems may offer digital competition.

Table 1.1 SWOT Analysis of the Proposed Digital Agriculture Collaboration and Support System (DACS)

The SWOT analysis identifies several strengths for the Digital Agriculture Collaboration and Support System (DACS). The integration of several components into one system is a strength of DACS. With DACS, integration of Customer Information Management and Payment Records, Seminar Management, the Certified Breeder Registry, Inquiry Management, Management of Historical Data, and Analytics and Reporting is possible. DACS will streamline the implementation of certain business rules for the organization, such as seminar prerequisites, breeder eligibility, and certification validity. DACS aids in the elimination of redundant records, the need for repeated data entry, and the time-consuming process of manual verification.

From the preliminary investigations and stakeholder interviews, the existing business processes of the organization are characterized by poor information management, inefficient handling of transactions, repeated manual verification, and time-consuming processes. DACS addresses the problems by consolidating the organization's business processes and creating a central repository for operational records. The Inquiry Management module provides a framework for the submission of tickets and the monitoring of ticket status, as well as the response and resolution of tickets.

The proposed system, despite the advantages, has some drawbacks that include an unreliable internet connection, the time and effort spent on migrating and validating historical data, the need for ongoing technical support, and the varying levels of digital literacy. The reliability of the proposed system and the information generated are directly proportionate to the quality of data that are migrated.

The rising implementation of digital technology in the Philippine agricultural sector could allow for further refinements of the system and greater potential for wide-scale organizational deployment. To counter potential threats like cyber security, data privacy, process automation resistance, poor internet access, and changing organizational needs, Dominant Asia Poultry Genetics would need to ensure appropriate controls, provide user training and retraining, keep timely operational logs, and perform routine system maintenance and updates. These would ensure the long-run effectiveness, reliability, and sustainability of DACS.

d. Project Testing and Evaluation

To maximize the quality, reliability, and effectiveness of DACS, the proponents will conduct a series of evaluations and tests at every phase of development and implementation. The study employs the Scrum framework, one of the Agile Software Development Methodologies. Here, an iterative and incremental design is practiced through the execution of Sprints. Each unit of design, known as a Sprint, will be preceded by a planning phase and will contain the development, evaluation, and refinement of the system. This allows the proponents to make ongoing improvements to the system based on the findings from evaluations and feedback from stakeholders.

The evaluations and tests will verify that DACS, as formulated, meets the requirements of the specified goals and objectives of the system, which include: Office operations systematization; streamlining and optimizing workflow; strengthening monitoring of queries and transactions; improving the experience of the users and customers of the system; and enhancement of management and organizational decisions. These evaluations will include, but will not be limited to, Comparative Analysis, Functional Testing, Integration Testing, User Acceptance Testing (UAT), and ISO/IEC 25010 Software Quality Assessment.

The methodologies of testing and evaluation chosen will pertain directly to the objectives of the project as stated in Section a. Description and Objectives.

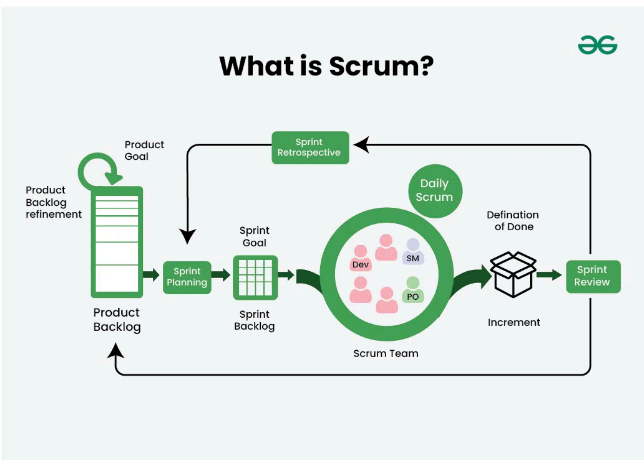

Fig. 1.6 Scrum Framework under the Agile Software Development Methodology

i. Respondents (Client + Customers)

This study identifies and groups two of the major stakeholders in the proposed Digital Agriculture Collaboration and Support System (DACS).

Qualitative respondents will be Dr. Erwin Joseph Cruz, Mrs. Cristina Cruz, the IT Staff, and some Dominant Asia administrative staff. These respondents are internal stakeholders of the organization. They will be involved in interviews and consultations, and will help the study understand the current operational processes and how the proposed system will help manage system and process requirements. They will also be involved in the system's design and help the study identify potential issues with the system. As internal stakeholders with access to the system, they will also review the proposed modules for Customer Information Management, Order Management and Payment Recording, Seminar Management, the Certified Breeder Registry, Inquiry Management, Historical Data Management, and Analytics and Reporting, and provide feedback on the modules' effectiveness and usability.

Quantitative respondents will be 30 registered and potential poultry farmers from 18 and older in the Dominant Asia network. To enhance the quality of the sample, the proposed study will employ purposive sampling methods. This will ensure that respondents have pertinent experience relating to the organization's system and services.

Farmer respondents, being the primary end users, will assess the system features for managing customer profiles, accessing seminars, monitoring seminar progress, submitting and tracking orders, uploading proofs of payment, and checking breeder eligibility and certification status, as well as submitting inquiry tickets, monitoring status, and accessing the FAQs Knowledge Base. Farmer respondents will take a pretest survey, use DACS for twenty-one (21) consecutive days, and take a posttest survey to assess the change in their user experience.

Farmer and internal respondents will fill out the User Evaluation Questionnaire. Part I will include the User Acceptance Testing Questionnaire, and Part II will include the ISO/IEC 25010 Software Quality Evaluation Questionnaire. Functional Testing and Integration Testing will be carried out by selected respondents only, based on their assigned roles, authorized access to the system, and the modules pertaining to their role. Responses to inquiries will be emailed, while the DACS will be assessed based on the ticket submission, recording, and status-functions.

ii. Instruments

To analyze the Digital Agriculture Collaboration and Support System (DACS) proposal, the research team will implement multiple instruments, testing procedures, and evaluative mechanisms to identify the extent to which integration, user acceptance, and the usability of the system are achieved. Based on a mixed-methods approach, the research instruments will gather both qualitative and quantitative data, which will answer the research questions concerning the extent to which DACS accomplishes its objectives and fulfills user requirements for all the integrated modules.

The Interview Guide was applied to obtain qualitative data from the internal stakeholders, Dr. Erwin Joseph Cruz, Mrs. Cristina Cruz, the IT Staff, and some of the Dominant Asia administrative personnel. The purpose of the interviews was to obtain information about the operational processes of the organization, the challenges that personnel and users face, and the requirements and feedback for the proposed system. Responses were recorded and analyzed through thematic analysis to determine system requirements.

The Pretest Questionnaire will be completed by thirty (30) registered and potential poultry farmers, prior to their use of DACS. This will determine their present user experiences with the operational processes of the organization and the services that the organization offers, such as submission of user information, participation in and the monitoring of seminars, submission and tracking of orders, submission of proof of payment, checking of Breeder eligibility and certification status, submission and tracking of inquiries, and the use of services offered by the organization.

After taking the pretest, users will engage with DACS for a period of twenty-one (21) days. They will access system functions related to their transactions (seminar attendance, submissions of orders and payments, checks of status as a breeder, tickets of inquiries, tickets of inquiries with statuses, and the FAQs Knowledge Base) as well as monitoring payments. Answers to inquiries and resolutions will continue to be sent via the organization's email channel.

The Posttest Questionnaire will be administered after the usage period of twenty-one (21) days. The posttest will capture variations in user experience once they have engaged with DACS. Pretest and posttest results will be analyzed to assess if the introduction of DACS had a positive impact on the experience of respondents interacting with the services of the organization, and if this was a change of a statistically significant nature.

1. **Comparative Analysis**

A Comparative Analysis will be carried out to assess the effectiveness of the proposed Digital Agriculture Collaboration and Support System (DACS) by considering the experiences reported by potential users and the experiences reported by the processes of some specific organizations.

For the quantitative assessment, thirty (30) currently registered and expected poultry farmers will answer a Pretest Questionnaire to assess DACS. This sample will then operate the proposed system for the next twenty-one (21) consecutive days and will answer a Posttest Questionnaire. The pretest and posttest data will be subjected to normality by the Shapiro–Wilk Test. If the data are normally distributed, a one-tailed, Paired-Sample t-Test will be conducted. If the assumption of normality is not satisfied, the alternative will be the Wilcoxon Signed-Rank Test. All statistical assessments will be carried out at a 0.05 threshold of significance.

The proponents also may compare the experiences reported by specific organizational processes that consist of certificate generation, updating a customer record, verification of a breeder's eligibility, and monitoring the status of inquiries, before and after the implementation of DACS. These processes will be evaluated under the same circumstances. The assessment will be used to analyze if DACS will shorten processing time, decrease repetitive manual tasks, and optimize the process.

Activity	Existing Process	DACS Process	Improvement	Interpretation
Generate Certificate	3 - 7 days	1 minute	90%	Very High Improvement
Update Customer Record	10 - 15 minutes	1 minute	80%	Very High Improvement
Verify Certificate	1 - 3 minutes	10 seconds	94%	Very High Improvement

Table 1.1 Comparative Analysis Template for Digital Agriculture Collaboration and Support System (DACS)

2. **Functional Testing**

Functional Testing will determine whether the features and modules of the proposed Digital Agriculture Collaboration and Support System (DACS) comply with the requested functional requirements and with the proposed system's operations. This kind of testing will establish whether the functions of the system provide the expected outputs for the defined test scenarios.

The assigned respondents will analyze the essential functions of DACS based on their roles as users and as holders of the authorizations to access the system. These functions include: User Authentication and Role-Based Access Control; Customer Information Management; Management of Orders and Record of Payments; Management of Seminars; Certified Breeder Registry; Management of Inquiries; Management of Historical Data; and Analytics and Reporting.

For each defined test case, the test scenario, expected result, actual result, and status will be recorded. If the actual result corresponds with the expected result, the test case will be defined as Passed. The test case will be considered Failed if the actual and expected results are not the same. If a test case was Failed, the noted issue, necessary corrective measure, and retest status will be recorded. The results of the Functional Testing will provide a basis as to whether the proposed system fulfills the functional requirements as defined.

Test Case ID	Module	Test Scenario	Expected Result	Actual Result	Status
FT-001	User Registration	Register a new farmer account	Farmer account is successfully created and stored in the database.		
FT-002	Seminar Management	Register a farmer for a seminar	Seminar registration is successfully recorded and reflected in the participant list.		
FT-003	Order Management	Submit a Parent Stock order	Order is successfully recorded and assigned the appropriate order status.		

Table 1.2 Functional Testing Template for Digital Agriculture Collaboration and Support System (DACS)

3. **Integration Testing**

Integration Testing will determine if the interconnected modules within the Digital Agriculture Collaboration and Support System (DACS) are able to properly interact with one another within the larger business workflows. DACS consolidates multiple constructs, and testing will ensure that information in one module is properly reflected in the associated modules.

Integration Testing is conducted on each module to determine the expected vs actual behavior of the system. If the expected behavior is met, the scenario will be marked as Pass. However, if the expected behavior is not met, the scenario will be marked as Fail. Failed scenarios are to address the issue that was witnessed and describe the action taken along with the status of the testing once the necessary actions were completed.

Test Case ID	Modules Involved	Integration Scenario	Expected Result	Status
IT-001	Seminar Management and Order Management	A farmer completes Seminar Modules 1, 2, and 3 and attempts to place a Parent Stock order.	The farmer's seminar completion is recognized, and Parent Stock ordering access is enabled.	
IT-002	Order Management and Payment Recording	A farmer uploads proof of payment for an existing order.	The payment record is linked to the correct order, and the order status is updated to indicate that payment has been submitted for verification.	
IT-003	Order Management and Certified Breeder Registry	A completed Parent Stock order reaches the required reference date for breeder eligibility.	The farmer's eligibility record is updated, and the ninety-day eligibility monitoring begins based on the completed Parent Stock transaction.	

Table 1.3 Integration Testing Template for Digital Agriculture Collaboration and Support System (DACS)

4. User Acceptance Testing

User Acceptance Testing (UAT) will be performed to identify whether the newly proposed Digital Agriculture Collaboration and Support System (DACS) will meet the requirements, expectations, and everyday needs of the end-users. After conducting the assigned Functional and Integration Tests, survey respondents will complete Part I of the User Evaluation Questionnaire to express their feedback on the proposed system. The UAT will focus on the thirty (30) registered and prospective poultry farmers after their completion of the Pretest Questionnaire, their use of the DACS for twenty-one (21) days, and their completion of the Posttest Questionnaire. Internal respondents will be the Owner/Executive, the IT Staff, and the Administrative Staff. These respondents will evaluate the system based on the modules and functions that are pertinent to their roles and that are within their access permissions.

UAT will focus on the system's perceived ease of use, the perceived ease of navigation and task completion, the perceived efficiency of workflow, overall user satisfaction, and the perceived usefulness. A four-point Likert scale will be used to collect survey responses, with the choice ranges of Strongly Disagree (1) to Strongly Agree (4). The responses will be evaluated to identify the concerns of users regarding the system's workflow and to suggest improvements for DACS prior to the system's final deployment. UAT of the Inquiry Management module will focus on the submission of Inquiry Tickets, access to the Frequently Asked Questions Knowledge Base, and monitoring of ticket statuses. Responses and resolutions to inquiries will continue to be conducted via email.

5. ISO25010 Evaluation

The second part of the User Evaluation Questionnaire will be based on the ISO/IEC 25010 Software Quality Assessment. This assessment will evaluate the perceived software quality of the Digital Agriculture Collaboration and Support System (DACS) and is based on the observable quality attributes of the system to the end users. The first version of this questionnaire will focus on Functional Suitability, Performance Efficiency, Reliability, and Usability.

Functional Suitability refers to the extent to which the DACS has all the functions needed for users to perform their tasks in DACS in the correct and complete manner. Performance efficiency will be considered in the context of the system's required response times and the system's required performance for routine actions under the expected normal conditions of use. Reliability will be assessed in terms of the system's performance, accuracy of record-keeping, and the system's fault tolerance and failure interruption. Usability will be assessed in terms of the internal users and also the registered and the expected poultry farmers ease of learning, navigating, and operating the system.

The statements will be rated based on a 4-point Likert scale from 1 (Strongly Disagree) to 4 (Strongly Agree). The perceived software quality of DACS for the given ISO/IEC 25010 metric dimensions will be determined using the weighted mean from the collected data.

Quality Characteristic	**Internal Users (Owner/Executive, Administrative Staff, and IT Staff)**	**Farmers**
Functional Suitability	Customer Information Management, Order Management and Payment Recording, Seminar Management, Certified Breeder Registry, Inquiry Management, Historical Data Management, Analytics, and Reporting	Account registration, customer-profile management, access to seminars, submission of orders, upload of proof of payment, order tracking, checking eligibility and certification status of breeders, submission of inquiry tickets, access to FAQs, and monitoring of ticket status
Performance Efficiency	Dashboard loading, report generation, payment verification, ticket retrieval and status update, and customer record management and editing	Login, upload of proof of payment, submission of inquiry ticket, retrieval of ticket status, access transaction history, and seminar content
Reliability	Accuracy and consistency of operational reports and records pertaining to customers, orders, payments, seminars, certifications, inquiries, and tickets along with historical records	Accuracy and consistency of seminar progress, payments, orders, inquiries, tickets, and certification and eligibility status of breeders
Usability	Dashboard navigation and management of customers and orders, along with participation in and administration of seminars, management of inquiries, ticket monitoring, and report generation	Registration and navigation of user profile, access to seminars, order and payment upload tracking, status checking, inquiry submission, access to FAQs, and monitoring ticket status

Table 1.4 Selected ISO/IEC 25010 Quality Characteristics Used for System Evaluation.

iii. Data Gathering Procedure

The first step in the data collection process was conducting semi-structured interviews with the Owner/Executive, Administrative Staff, IT Staff, and some of the registered poultry farmers at Dominant Asia Poultry Genetics. These interviews aimed to analyze and understand the operational processes, challenges, and needs of the organization in addition to identifying the gaps and requirements in the existing systems. Interview responses were transcribed and subjected to thematic analysis. The resulting themes were used to identify the requirements for and to design the Digital Agriculture Collaboration and Support System (DACS).

Once the system has been built, 30 registered and interested poultry farmers will be selected to participate in a system evaluation. Participants will answer a Pretest Questionnaire to establish their prior level of engagement and interaction with the prevailing system and organizational services, as well as the service and process accessibility for the following: seminar participation, submission of customer information, order and payment processing, breeder certification, inquiry follow-ups, and general services.

The proponents will lead an orientation to assist respondents in understanding the workflows of DACS and DACS features after the pretest and before the 21-day system trial. During the 21-day trial, respondents will perform DACS relevant organizational transactions, which include but are not limited to, accessing seminar modules and monitoring progress, submitting and tracking requests, uploading payment proofs, checking breeder eligibility and certification status, submitting and monitoring inquiry tickets, and accessing the FAQs Knowledge Base. DACS will record and monitor ticket status. Responses to inquiries and ticket resolutions will be provided via email to the organization, while DACS will track ticket status.

The proponents will review DACS usage during the 21-day trial using DACS activity logs. The logs will only be reviewed to confirm that respondents participated and used the system. Respondents will be notified about the purpose of the logs, and the logs will be used only for the purpose stated and will comply with data privacy laws.

After the 21-day trial, respondents will complete a Posttest, User Evaluation (User Acceptance Testing), and ISO/IEC 25010 Software Quality Evaluation (Usability Test). Other participating respondents will complete Functional Testing and Integration Testing based on their assigned user role.

The gathered qualitative and quantitative data will be analyzed after being integrated. Thematic analysis will be used for the interviews, and pre/post test results will be analyzed using the relevant descriptive and inferential statistics as outlined in the study. The results from Comparative Analysis, Functional Testing, Integration Testing, User Acceptance Testing, and the ISO/IEC 25010 Software Quality Evaluation will be used to assess the effectiveness and acceptability of DACS, along with the perceived software quality.

iv. Analysis

The analysis of qualitative and quantitative data from various tests and evaluation methods will assess the proposed Digital Agriculture Collaboration and Support System (DACS) for its functionality, usability, reliability, and effectiveness, as well as the overall acceptability of the system. This study utilizes a mixed methods research design. The quantitative component will be used to assess the changes in user experience, user acceptance and system functionality and Integration as well as the perceived software quality while the qualitative component will assist in detailing the operational and implementation challenges of the organization and the stakeholder's needs.

For the quantitative analysis, frequency, percentage, mean, and standard deviation will be used as descriptive statistics to summarize the demographic data and questionnaire data. The Shapiro-Wilk Test will be used to evaluate the assumption of normality for the difference score of the Pretest and Posttest Questionnaires. A paired sample t-test will be used to assess the difference of the mean scores if the difference scores of the Pretest and Posttest Questionnaires are normally distributed. A one-tailed test will be used in this case because the intention of this study is to assess the improvement of the user experience of the respondents after using DACS for a period of twenty-one (21) consecutive days. If the difference scores are not normally distributed, the Wilcoxon Signed Rank Test will be used. The 0.05 level of significance ($\alpha = 0.05$) will be observed in all the statistical tests that will be conducted.

Decision Rule. For a given significance level of 0.05 ($\alpha = 0.05$), the null hypothesis (H_0) will be rejected when the p-value is less than 0.05. In this case, the alternative hypothesis (H_1) will be accepted. If the p-value is greater than or equal to 0.05, the null hypothesis will be accepted.

For Functional Testing and Integration Testing, results will be analyzed using a Pass or Fail criterion. Evaluators will check if the actual output of the system matches the expected output for each test case. If the actual output meets the requirement, the test case will be marked as Pass. If the actual output does not meet the expected requirement, the test case will be marked as Fail. For test cases marked as Fail, the observed problem, the action taken to fix the problem, and the status of the test case (e.g., retested) will be included. The percentage of test cases that passed will be calculated as shown below.

PASS RATE = (Passed Test Cases / Total Test Cases) x 100

For User Acceptance Testing (UAT), the evaluation of DACS acceptance will cover the following measures for ease of use and ease of navigation, satisfaction achieved through task and workflow efficiency, perceived usefulness, and satisfaction and perceived readiness for implementation. Evaluation of DACS through the ISO/IEC 25010 Software Quality Evaluation will cover perceived formal satisfaction of DACS through Functional Suitability, Performance Efficiency, Reliability, and Usability. The results of the evaluations will be interpreted using the weighted mean, and the results will be analyzed based on the scales in Table 1.5.

For the qualitative analysis, thematic analysis will be conducted to interpret the results of the evaluation of the Owner/Executive, the Administrative and the IT Staff, and the selected registered poultry farmers. Based on the operational processes, obstacles and needs in functional processes, as well as the system and implementation, significant statements will be captured and coded. The related codes will be clustered to form operational categories to outline the recurrent themes. The themes will provide descriptions of the stakeholder's experiences and will be in support of the quantitative findings.

The weighted mean will be computed using the following formula:

$$\overline{X} = \frac{\Sigma f x}{N}$$

- $\overline{X}$ **:** The Weighted Mean
- f **:** Frequency of responses for each rating
- x **:** Numerical Value Assigned to the Response
- N **:** Total Number of Responses

Mean Range	Interpretation
3.26 - 4.00	Excellent (Strongly Agree)
2.51 - 3.25	Good (Agree)
1.76 - 2.50	Fair (Disagree)
1.00 - 1.75	Poor (Strongly Disagree)

Table 1.5 Interpretation of Mean Ratings

The interpretation of the calculated weighted mean for every UAT criterion and every ISO/IEC 25010 quality characteristic will be according to the scale shown in Table 1.5. These results and the results of the pretest and posttest comparison, Functional Testing, Integration Testing, Comparative Analysis and the Thematic Analysis will be combined to provide an overall assessment for the DACS regarding the effectiveness, acceptability and the perceived software quality of the system.

CHAPTER 2 : REVIEW OF RELATED WORKS

This chapter describes the current systems, applications, and digital solutions pertaining to poultry management, breeder management, agricultural information systems, farmer training, certification management, and order management. The systems and applications are described in terms of their features, strengths, weaknesses, and their relevance to the Digital Agriculture Collaboration and Support System (DACS). The proponents of this system review aim to evaluate the available technologies to identify the gaps and defend the need for the proposed system.

Enterprise Solution for a Poultry Management System

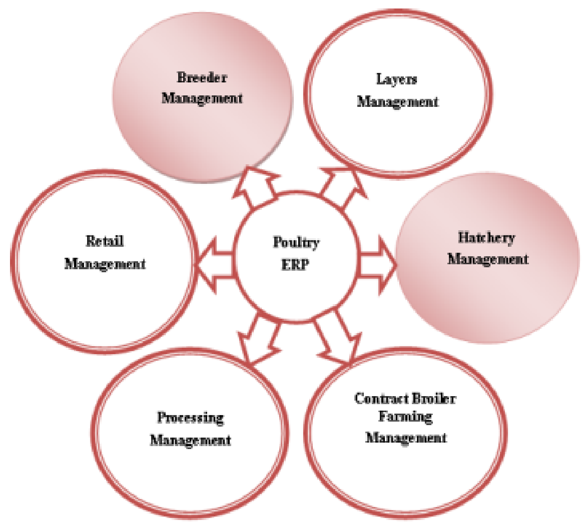

Fig. 2.1 Poultry ERP module

Geetha et al. (2020) introduced a business application for poultry management. It provides process integration for poultry management within the information systems business framework. It explained the advantages of integrated systems for management and data collection concerning redundancy and information consistency within the business frame. Integration also enhances the efficiency of poultry management in the organization.

The research described how poultry operations that are managed manually are prone to duplication of records, disintegration of data, and inefficient operations. Geetha et al. (2020) further noted that integrated enterprise solutions provide poultry management with the ability to access information, and manage business processes and decisions with data provided through an integrated system. They also argued that enterprise integration at the poultry management level reduces the inefficiencies of systems and the inconsistency of information required to conduct daily poultry management operations.

Poultry Farming Information Management System

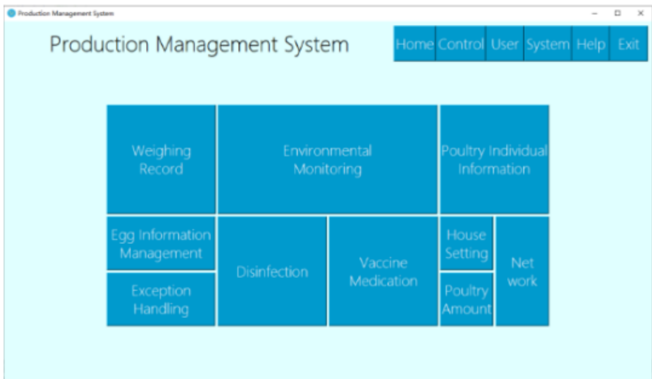

Fig. 2.2 Production Management System

The Poultry Farming Information Management System developed by Zheng et al. (2021) offers a digital solution to the problem of recording, storing, and accessing data from poultry farms by creating a centralized system. To resolve the problems posed by recording data manually, this system provides a framework for the management of production data, stock, and other operational data pertaining to poultry. Zheng et al. (2021) argue that information management systems assist poultry companies in the accurate recording of data, the elimination of redundancy, and the quick access of operational data.

Centralized poultry data management was proven to enhance the quality of the decisions made in the poultry industry and to increase the efficiency of the operations. Zheng et al. (2021) reported that use of cloud-based and database systems provides system users the capability to access and update records in real time. The collaboration among the units of the organization improves, as does the consistency of the information. The system also showed the reduction of the administrative burden and the manual data entry errors due to the system.

Layer Poultry Farm Management System

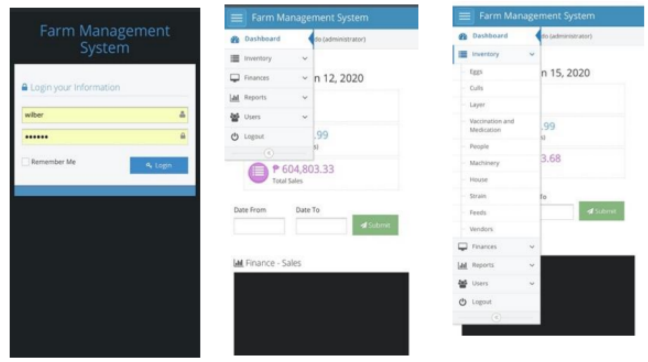

Fig. 2.3 Layer Poultry Management System

To digitize poultry farm operations and records, Sabado (2024) created the Layer Poultry Farm Management System. Sabado (2024) claims that the digital management system eliminates issues caused by manual record keeping and helps with the administration of farms. The system helps farm operators manage production processes and keep records and helps improve the accessibility of information pertaining to operation activities.

The system being studied allows poultry farm operators to manage farm records, farm inventory records, and farm production records. Sabado (2024) also states that the potential system users rated the system as acceptable in its design and ease of use, thus allowing the operators of poultry farms to have a better functionality of farm operations. The centralized information storage drastically reduced the chances of information loss or duplication and also enabled rapid information retrieval.

Breeding Management Information System

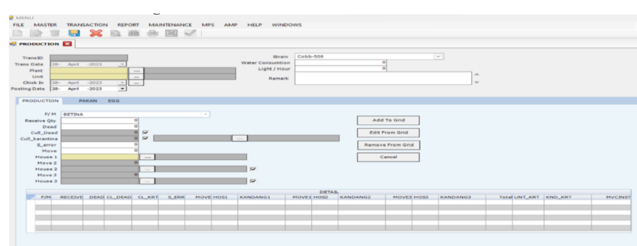

Fig. 2.4 Breeding Management Information System

Gunawan and Kuswanto (2024)'s Breeding

Management Information System helps automate the management of breeding data and operational records in livestock and poultry breeding. It helps organizations capture, monitor, and retrieve breeding information. Thus, organizations can collect and keep breeding records in an orderly and timely fashion. Gunawan and Kuswanto (2024) describe the centralized breeding information system as a means to achieve greater data accuracy, faster breeding records, and improved breeding operation management.

The study focused on the need for extensive breeding records for operational and management decision support. It was noted that the system helped the management of breeding-related data, and the system reduced the chances of data inconsistency. Breeding records digitization helped create a more reliable and formal system of managing breeding activities.

Poultry Records Monitoring System

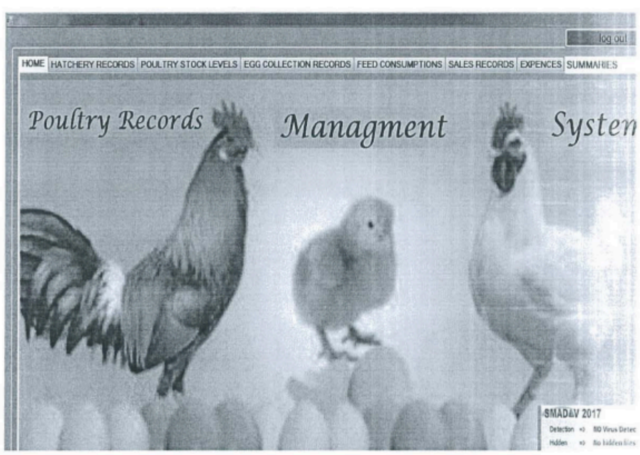

Fig. 2.5 Poultry Records Management System

Isaac (2017) published a study on a Poultry Records Monitoring System, an automated way to handle, track, and report poultry records. The creator developed the system to offer a solution to the problem of manual record keeping by offering a system to store and keep track of poultry information. Isaac (2017) reports that manually keeping poultry records is labor intensive and that poultry record digitization helps make operational tasks easier.

Isaac (2017) recommends poultry digitization for the decision-making process on the management of the farm and helps reduce the labor of record keeping. The author reports that the system helps keep track of poultry breeders' information, helps monitor operational task records, and helps formulate reports of poultry and management task performance. The Poultry Records Monitoring System centralized the records and reduced the risk of information being lost and helped improve information retrieval.

Mobile Application for Native Chicken Management

Fig. 2.6 Thai Native Chicken mobile app

Saengwong and Koksantia (2020) introduced the Mobile Application for Native Chicken Management to provide farmers with educational resources and management services through mobile technology. Farmers are able to easily access management tools and information on native chickens through the application. As stated by Saengwong and Koksantia (2020), mobile learning systems improve knowledge and educational services for poultry farmers.

This study is an excellent example of the application of mobile technology for farmer education and knowledge transfer. As indicated by Saengwong and Koksantia (2020), the application improved access to poultry management information and created a convenient learning environment that is accessible at any time and at any place. The system also made it possible to use mobile technology to advance agricultural education and tailor it to users who have limited access to other training and education services.

Web Portal Poultry Farming System with Real-Time Analytics

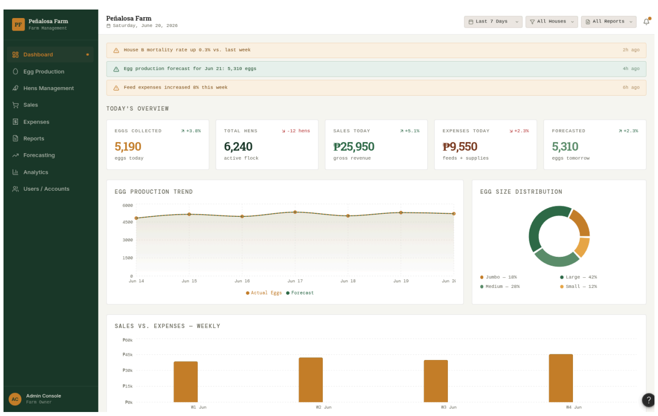

Fig. 2.7 Sample reconstructed interface of the Web-Portal Poultry Farming System with Real-Time Descriptive Analytics

Leovin et al. (2026) introduced Web Portal Poultry Farming System with Real-Time Analytics which combines data analytics and reporting and is used to optimize the management of poultry farms. The authors developed a system with dashboards, analytical tools, and real-time reporting to assist users with monitoring, evaluation, and proactive decision making, among others. Leovin et al. (2026) stated that organizations could gain insights to support and improve their planning and management processes by utilizing an analytics-centric system.

This research also stated that data visualization and real-time reporting aid data-driven decision making. The authors also stated that the dashboard of the system was useful for the users to efficiently view pertinent information, analyze the data to find operational trends and patterns, and monitor daily and/or periodic key operational performance indicators. Furthermore, the availability of reports that were updated in real-time also enhanced the organization's ability to address operational issues and business challenges promptly and make data-driven decisions in a timely manner.

Web-Based Platform for Poultry Market Linkages

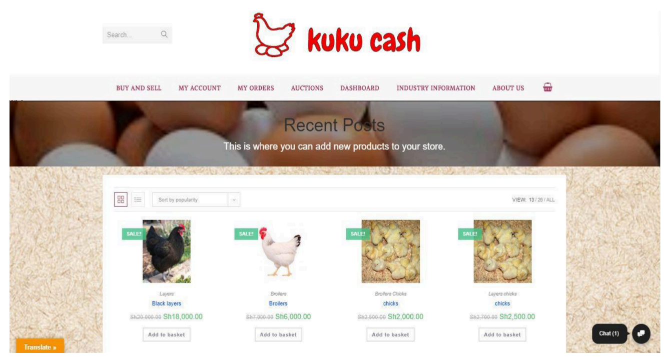

Fig. 2.8 Kuku Cash

Mambile et al. (2018) proposed the first Web-Based Poultry Market Linkages Platform. Customers and Poultry Producers communicate using the digital platform. Mambile et al. (2018) cite that the digital platform improves communication between the stakeholders and improves the market. In addition, the digital platform improves the accessibility and the efficient workflow of communication between the stakeholders. Mambile et al. (2018) state that the digital market platforms improve the presence and the involvement of the customers through the digital platforms that allow customers to ask about the products and the platforms that allow customers to post queries about the products and facilitate the transactions.

The study described the role of web-based systems to improve the customers' experience and accessibility to a service or market in the context of an agricultural market system. Mambile et al. (2018) explained that the system made access to market information, and the ability to communicate to buyers/sellers, and engage in poultry-related business activities, much easier. The system essentially revealed that the use of web-based systems and services offers market participants greater flexibility in communication while reducing system-related constraints associated with conventional market practices.

CHAPTER 3 : DESIGN METHODOLOGY

This chapter describes the design methodology applied to the Digital Agriculture Collaboration and Support System (DACS). It outlines the concepts of the design, the logic behind the project, the design, and the framework of the system, and the design requirements of the system. It also describes how the operational challenges, needs and wants of system users, the existing workflows and the business operating procedures of Dominant Asia Poultry Genetics, were converted to the functional, non-functional, and user requirements of DACS.

The project employs the Scrum framework of Agile Software Development Methodology. With the use of iterative and incremental Sprints, the project team will conduct the tasks of requirements analysis, and design and develop the system, and conduct the tasks of system testing, reviewing and refining, and so on during the course of the project. DACS aims to provide a robust, easy to maintain, scalable, secure, easy to use web and mobile application that includes modules on User Management with Role-Based Access Control, Customer Management with Order and Payment modules, Seminar Management, the Certified Breeder Module, Inquiry Management, along with Historical Data and Reporting with Analysis. Within DACS, users' inquiries will be logged and tracked. Responses and resolutions to users' inquiries will be conducted and sent via the company's official email.

A. Conceptual Framework

In this stage of the framework, we specify the information, requirements, and resources needed for the design of the Digital Agriculture Collaboration and Support System (DACS). The input consists of operational issues captured from the interviews and consultations with the stakeholders that include inadequate management of data, inefficient operational activities, disjointed systems, and low levels of customer satisfaction. This stage also includes the functional and non-functional requirements and user requirements of Dominant Asia Poultry Genetics, together with the results of the pretest, and the available records of the organization. This input is vital for the design, development, and assessment of the system.

In this stage, we detail all the activities and procedures for the design, development, implementation, and evaluation of the proposed system. The development of DACS will employ the Scrum Agile method, which will allow the iterative development of the system through planning, design, development, testing, and continuous improvement of the system, based on the feedback of the stakeholders. After development of the system, the system will be evaluated through Functional, Integration, and User Acceptance Testing, and the assessment of the system through the ISO/IEC 25010 Software Quality. To assess the system's effectiveness, a pretest-posttest evaluation will be conducted. For this evaluation, the participants will use the system during the intervention, and will then complete the posttest questionnaire. The quantitative and qualitative data collected for this evaluation will be analyzed through the relevant statistical and thematic analyses to assess the effectiveness of the system and the extent to which the system meets its goals and objectives.

The Output stage illustrates the anticipated outcomes from the implementation of the Digital Agriculture Collaboration and Support System (DACS). This system is expected to create a dedicated, integrated, and automated platform for the management Breeder, Farmer, Order, Payment, Inquiry, Consultancy, and Business Analytics. DACS is expected to bring improvements to data management, operational inefficiencies, system fragmentation, user experience, and management reporting and decision-making through integrated and automated process management. The evaluation findings will likely capture the system's functional performance, qualitative perspective on the software, and overall satisfaction of the users.

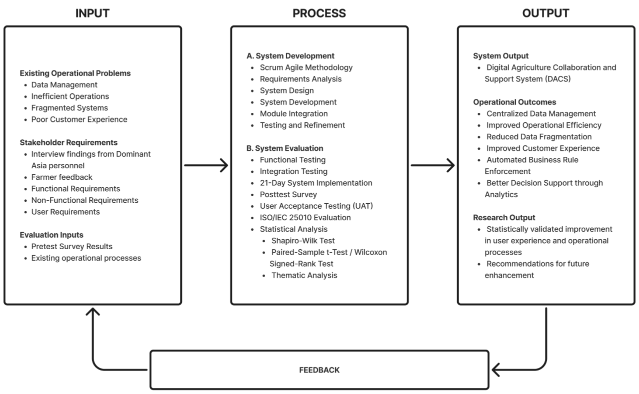

Figure 3.1 Conceptual Framework of the Study

The Input-Process-Output (IPO) model, seen in Figure 3.1, outlines the methodology of this research. In this model, the issues within the organization, the needs of the stakeholders, and the study data, constitute the inputs for the design of the Digital Agriculture Collaboration and Support System (DACS). The Scrum Agile model of software development coupled with the system's evaluation, will transform the stated inputs into a unified online system, from which Dominant Asia Poultry Genetics stands to gain a variety of operational improvements. Among the anticipated operational benefits will be the development of a system to manage data and business operations that will be efficient and effective and that will improve customer satisfaction. The system will be a business operations information system with an adequate and appropriate operational and software quality.

B. Project Timeline

The following outlines the expected sequence of project activities for the Digital Agriculture Collaboration and Support System (DACS) project. Activities include the collection and analysis of system requirements; design, development, and iterative refinement of the system; testing and evaluation of the system; and documentation of the system. This project employs the Scrum methodology of the Agile Software Development Method. This feature of the DACS project allows the proponents the flexibility to iterative and incrementally refine the system by conducting development, testing, and review activities within the Sprints.

In the month of August, and extending into the month of September, the proponents will perform the primary development and integration of the following key DACS modules: User Management; Customer Information Management; Payment and Order Management; Seminar Management; Certified Breeder Registry; Inquiry Management; Historical Data Management; and Analytics and Reporting Modules. The Inquiry Management Module will aid the submission, recording, and status tracking of inquiry tickets. The responses and resolutions to tickets will be relayed via the organization's email channel.

During the month of October, DACS will complete a battery of testing. These will include, but are not limited to, Functional Testing, Integration Testing, Comparative Analysis, and User Acceptance Testing (UAT). Assessment of DACS against the ISO/IEC 25010 Software Quality Standards will also be conducted. Testing and evaluation will be carried out by authorized personnel of Dominant Asia Poultry Genetics and a selected sample of registered and prospective poultry farmers. Results of the testing will be used to frame and prioritize system issues through the assessment of the user experience and the perceived quality of the proposed system.

During the month of November, the project team intends to modify and improve DACS. This will incorporate testing and evaluation findings, feedback, and concerns. Changes will address the system's functionality and reliability. Usability, performance efficiency, and user experience will be incorporated as well.

The project team intends to complete the final adjustments to the system and conduct a consolidation and analysis of the testing and evaluation results. This will be accompanied by the finalization of project documentation and the preparation of the system and research presentation.

The project timetable and phased development schedule for the Digital Agriculture

Collaboration and Support System (DACS) are presented in Figure 3.

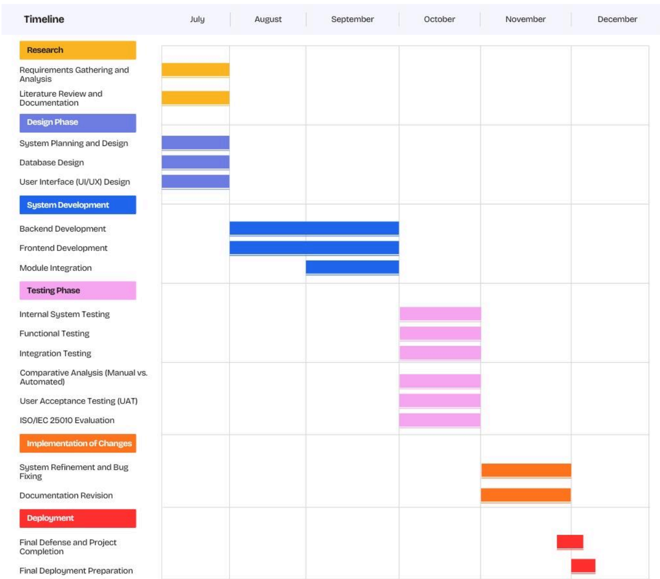

Figure 3.2 Project Timeline (Gantt Chart) for the Development of the Digital Agriculture Collaboration and Support System (DACS)

C. System Scope

DACS will replace the organization's piecemeal use of various formats and communications methods, such as Google Forms and Sheets, Facebook Messenger, standalone emails, and other unintegrated platforms. With DACS, operational records and workflows will be documented and managed in one, integrated system. DACS will reduce the time-consuming and redundant data entry and verification tasks. DACS will also provide poultry farmers and potential poultry farmers with more organizational services through a single user account.

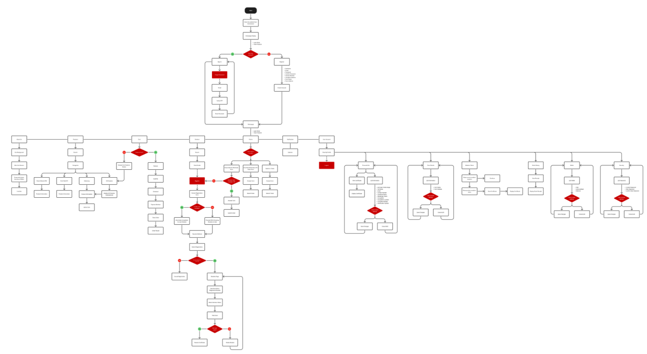

Figure 3.3 Customer Website System Framework

DACS is a proprietary, web-based, and mobile system developed for Dominant Asia Poultry Genetics. DACS integrates the various operation methods of the organization and focuses on the improvement of record management and the manual verification process. Furthermore, DACS provides the farmer clientele with a single user account for ease of access to the various services of the organization.

The system is built out of several modules including User and Customer Management, Order and Payment Management, Seminar Management, Certified Breeder Registry, Inquiry Management, Historical Data, and Analytics and Reporting. The system also contains a knowledge base, ticket status monitoring, role dashboards, operation record searches, and operational reports. DACS records the status of tickets for inquiries that are responded to and resolved via email.

The proposed system consists of the following core operational modules:

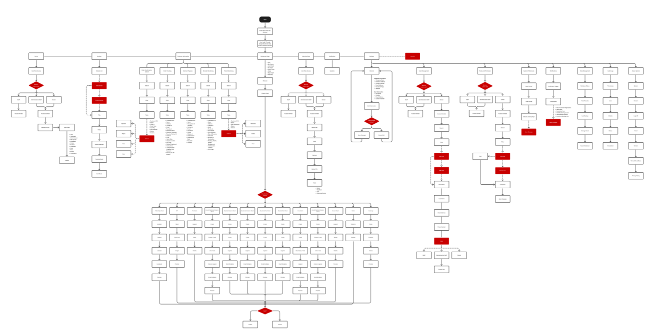

Figure 3.4 DACS System Framework

Customer Information Management

This module consolidates customer and farm data into a single database system. By centralizing this data, it can be utilized for multiple purposes, including Seminar registration, online orders, payment recording, inquiries, and Breeder certification for the user. Administrative Staff can search, view, and edit customer data, while clients/farmers can view and edit their permitted profile data.

Order Management and Payment Recording

This module allows registered clients/farmers to make online orders for Parent Stocks, F1 products and Veterinary products. Users can upload their proof of payment for the system to verify, and view the status of their orders and payment history. Administrative Staff can view the proof of payment and update the status for payment and the related order.

Seminar Management

This module manages the registration and recording of attendance and the tracking of learning progress, quizzes and completion for Seminars. The system enforces the business rule of the organization that mandates that farmers must first complete Seminars 1, 2, and 3, before being allowed to place orders for Parent Stocks.

Certified Breeder Registry

This module contains the central records of Breeder eligibility and Breeder Certification and the status and compliance to Qualification Certification. It assists the Administrative Staff in qualifying Breeders and applying the organization's rules of eligibility and validity. Farmers can view their Breeder eligibility and Breeder Certification status in their account.

Inquiry Management

Farmers can use the Inquiry Management module to submit questions they have concerning the offerings of Dominant Asia Poultry Genetics (i.e. products, seminars, breeder certification, etc.). Each inquiry is assigned a ticket number that indicates the current status of the ticket (e.g. Open, Closed) and is easily referenced by any Administrative Staff member or the farmer.

A Frequently Asked Questions Knowledge Base helps Authorized Staff with information that is commonly inquired about, and about which they will manage tickets. Inquiry tickets are recorded and monitored in DACS. The submission status of inquiry tickets (e.g. Submitted, Under Review, Responded, Closed) is also managed in DACS. Responses to inquiries are sent through the official email channel of the organization.

Historical Data Management

Records of customers, inquiries, and certifications are consolidated into an organized data module. Historical data is important for the organization and for its customers to have easily accessible information about past transactions and interactions.

Operational Reporting and User Management

Data from DACS is used to manage reports concerning the status of customer records, transactions, payments, seminar enrollments, certifications, and inquiries. Authorized Administrative Staff and the Owner/Executive have access to reports and data that enable them to make informed decisions about operations.

DACS is an online platform and therefore relies on an uninterrupted Internet connection.

The study focuses only the operations of Dominant Asia Poultry Genetics and will not cover:

- The functions of Enterprise Resource Planning, including modules for accounting, payroll, procurement, and inventory, as well as modules for logistics and supply chain management;
- Integrated online payment systems and real-time electronic payment processing;
- Functionalities of offline systems;
- Integrations of Third-Party Learning Management Systems; and
- Developments for other organizations outside Dominant Asia Poultry Genetics without

additional customization, verification, and validation.

D. Requirements Specification

This section collects the aforementioned requirements, including the functional, non-functional, and user requirements of the Digital Agriculture Collaboration and Support System (DACS). Given the nature of the system's support to Dominant Asia Poultry Genetics, the requirements define the essential capabilities, qualitative attributes, and features of the system.

The requirements were gathered from a series of semi-structured interviews, process mapping, a review of relevant documents, and the observation of organizational workflows and operations. The operational constraints and problems, the stakeholders' interests, and the objectives and boundaries of the system and the project (which were outlined in the previous sections) were considered in gathering the requirements.

The requirements outlined here include the minimum additional features that the DACS Minimum Viable Product (MVP) may include. These features include User Management and Role Based Access Control, Management of Customer Records, Orders and Payments, Management of Seminars, the Certified Breeder System, Management of Inquiries, Management of Historical Records, and Data Analytics and Reporting Services. These requirements will guide the design and development of the system and will describe the boundaries of the functional, integration, and user acceptance and evaluation tests of the software.

a. Functional Requirements

User Authentication and Role-Based Access Control

The system shall provide a secure authentication mechanism for registered users via Firebase Authentication, and shall provide access to system features and information by implementing Role-Based Access Control. The four user roles shall be Owner/Executive, IT Staff, Administrative Staff, and Clients/Farmers.

The Owner/Executive shall have full access to user account management, operational dashboards, reports, and system modules to facilitate organizational oversight and decision-making. Information Technology Staff shall be responsible for the system's configurations, settings, user access, and maintenance, and shall provide support and assistance whenever required.

Administrative Staff shall be responsible for the management of customer records, order processing, verification of uploaded proof of payment, monitoring of seminars, maintaining records of breeders' certification and eligibility, managing inquiry tickets, updating ticket statuses, and managing the FAQs Knowledge Base. Historical records and operational reports shall be the responsibility of the Administrative Staff. Actual responses and resolutions to inquiry tickets shall be provided to the Clients/Farmers via the designated email channel of the organization. The associated records and ticket statuses shall be retained in DACS.

Clients/Farmers shall have the ability to create and manage their own accounts, maintain their customer and farm information, access seminar modules, track their seminar progress, place and track orders, upload proof of payment, review their transactions, submit and track inquiry tickets, access the FAQs Knowledge Base, receive inquiry responses via email, and view their certification and eligibility status for breeding.

Users shall only have access to the system functions, records, and information necessary to perform the duties of their assigned role. The system shall restrict access to other modules, records, and administrative functions to unauthorized users in support of data security, confidentiality, and proper access control.

Online Ordering and Payment Recording

Registered clients/farmers will have the ability to order Parent Stock and F1 and veterinary products through the platform. This system will assist the organization in enforcing its business rule that Client/Farmers must complete Seminar Modules 1, 2 and 3 before they will be able to place a Parent Stock order.

The system will allow client/farmers to provide proof of payment for order payment verification, which will be done by the admin staff. The admin staff will have the ability to review proof of payment and will be able to set the payment and order status to verified and fulfilled, respectively. The system will show the client/farmer and the authorized personnel the current order status using the defined status of Pending, Payment Submitted, Payment Verified, Processing, Fulfilled and other status the organization may create.

The system will create a system notification or confirmation for the client/farmer for the successful order submission and for payment and order status update. For every registered client/farmer, the system will create and maintain a history of every transaction, including the order, payment, and status updates.

Certified Breeder Registry

The system will have a record of all of the certified poultry breeders and will be organized by region and province. For each record, the system will include a farmer's eligibility to be a breeder, certification status, date of certification, the period of certification, and Parent Stock transactions.

The system will have a record of the released date of a farmer's Parent Stock order and will use this date to determine the eligibility of the farmer to be a certified breeder. The system will enforce the organization's requirement of a ninety-day eligibility period by using the released date of the farmer's Parent Stock order.

The system will monitor the expiration of breeder certifications based on the date breeders were awarded seminar certificates. The certification will be valid for two (2) years per the policy of the organization. Per the relevant eligibility and expiration criteria, the system will determine the breeder's status as Active, Pending, Expired, or Ineligible.

The system will create a reminder for Administrative Staff about the recertification needs of the breeders. The system will allow users to view the breeders' records and status, and certification expiration based on criteria selection and filtration of records. This will aid the staff in the management of certified breeders.

Farmer Education and Seminar Module

The system will provide digital organization and management of the seminar of Dominant Asia Poultry Genetics. Seminar Modules 1, 2, and 3 will be embedded in the DACS to provide seminar participants access to the seminar, learning materials, and view progress of both the module and video. Seminar participants will be able to complete an automated quiz, and the system will provide the results.

The organization will enforce its rule of business by preventing Clients/Farmers from ordering Parent Stock prior to the completion of Seminar Modules 1, 2, and 3. Completion of the quiz and participant progress and completion of the seminar will be certified.

Eligible participants may request a Seminar Certificate of Attendance after completing the seminar modules. Certified participants may be verified or audited by the Administrative Staff. Records of seminar participants, course modules, and completion status, quizzes, and attendance, as well as requests for seminar attendance certificates, will be retained by the system for administrative and management purposes.

Inquiry Management

An Inquiry Management Module will allow Clients and Farmers to submit inquiries about the products, seminar, and breeder certification, as well as innovate orders, payments, and services that Dominant Asia Poultry Genetics offers. Submitted inquiries will be converted to an inquiry ticket and will have a reference number and status, which will allow the Clients and Farmers, as well as the Administrative Staff, to monitor the ticket.

To minimize repetitive inquiries and provide instant access to information for users, the system will contain a Frequently Asked Questions Knowledge Base. Authorized Administrative Staff will be able to monitor inquiry tickets.

Administrative Staff will be able to categorize and define the status of an inquiry ticket as Submitted, Under Review, Responded, and Closed. The responded and resolved inquiry will be communicated to the Client or Farmer through the email of the organization. DACS will retain a comprehensive record of inquires, reference numbers, personnel assignments, categories, statuses, and other date- and time-related monitoring information. Real-time messaging and the resolution of inquiries will not be conducted through the system.

Historical Data Migration

The system shall have the capacity to import, consolidate, and migrate existing operational records from Google Forms and Google Sheets. Records shall include customer data, transactions, and payments, as well as records of seminar enrollment and participation, inquiries, eligibility and certification of breeders.

The imported records will be saved into a centralized PostgreSQL database on Alibaba Cloud. The system will safeguard the accuracy, completeness, and integrity of the data with the least possible amount of migrated data duplication and inconsistency. Authorized Users will have the ability to search and filter, retrieve, and review organized historical data for customer history, operational, and management record review, verification, and report support.

The system will document the outcomes of the migration (results of the migration), which will include records that were successfully imported, records that were duplicates, were incomplete, or require manual review and/or correction.

Analytics and Reporting Dashboard

The system shall include an interactive Analytics and Reporting Dashboard that displays essential data for analysis and review of operational performance, for example, summaries of customer records, total monthly transactions by product, payment verification, seminar enrollment and participation records, qualification of breeders by region or province, and status and volume of inquiries and orders.

The various displays will facilitate management to view, analyze, and make decisions on system data and records. Authorized Administrative Staff, the Owner, and/or the Executive will have the ability to apply filters, view data summaries or details, and create or export operational reports formatted as required.

Access to dashboards, reports, and export features will be controlled based on the user's role and granted permissions.

b. Non-Functional Requirements

Usability

The interface of the system will be designed with the user in mind, especially the less technical user such as poultry farmers. The interface will be simple and easy to use on devices of all sizes from smartphones to tablets and PCs. Clear markings, recognizable icons, and consistency of design with text and buttons at the right size will be incorporated in the design of the interface.

Typically, users can perform tasks from the main dashboard with three (3) or less taps or clicks. The system will inform users of omissions or errors in the information they provide and will offer corrective assistance and guidance.

To minimize the learning curve for first-time users, the system will include an onboarding guide, contextual tips, and a FAQ Knowledge Base with tips on system usage. Slight variations in the design of the interface may occur, but the layout and navigational structure across the system will remain consistent.

Performance

The system will strive to meet a three (3) to five (5) second response time for the majority of system users under typical conditions and for standard load and mobile and broadband data usage. The system will maintain these thresholds even with a large number of users accessing the system.

Key system functions will have their response times tested for performance evaluation. Functions include user authentication, retrieval of customer records, access to seminar modules, submission of orders, uploads of proof of payment, submission of inquiry tickets, retrieval of ticket status, searches of historical records, loading dashboards, and generation of reports.

The system will reduce loading delays by optimizing database queries, page resources, file uploads, and data retrieval. For requests that the system anticipates will take longer to process, the system will track the request and indicate to the user that processing is taking place.

Reliability and Availability

The system will function reliably under typical use. The system will minimize unanticipated errors through validation and error handling. The system will ensure erroneous records will not be created for customer records, orders, payment records, seminar records, breeder certification records, inquiry tickets, or other operational data.

Temporary loss of network connectivity will not cause the system to cease functioning. The system will minimize unexpected interruptions of service by properly validating and handling errors. Users will be allowed to continue interrupted operations with the least amount of data loss. The system will avoid creating multiple records from duplicate requests, and will acknowledge the successful submission of records.

The system will provide consistency between linked modules and not allow any incomplete or invalid transactions to be logged as successful. Backups and recovery of the database will be routinely scheduled to minimize the permanent loss of data and to assist in the recovery of business data after an unanticipated data loss and/or system recovery.

Security

The system will utilize Role-Based Access Control (RBAC) to ensure users will have access to only the portions of the system, records, and data allowed by their role. Access and the management of user sessions will be accomplished by Firebase Authentication for enforced secure access and verification.

All operational data will be stored in a centralized PostgreSQL database with Ali Baba Cloud. This data will include customer records, order data, records for proof of payment and seminars, breeder data and their eligibility and certification, inquiry tickets, data for history and reporting, and all other operational data. Access to the database will be secured.

User passwords and other sensitive data will not be stored in the DACS operational database. The system will validate user input and file uploads to mitigate the risk of system compromise, data submissions, and access of a malfeasive nature. Access to system functions, reports, records, payment data, and personal data of customers will be restricted to authorized users only.

Inquiries will be responded to and resolved using the designated email of the organization. For the purposes of centralized tracking, DACS will only retain the details of the inquiry, ticket number, classification, the employee to whom the inquiry was assigned, the status, the pertinent dates, and all other monitoring data.

The system should ensure the protection of personal and operational data from unauthorized access and other threats. The collection, processing, storage, and use of personal data must be in accordance with the provisions of the Data Privacy Act of 2012 or Republic Act No. 10173, and other data privacy and security compliance provisions.

Maintainability

The system shall use a modular design to facilitate the future maintenance and the modular design shall be used to facilitate the future enhancement, testing, and debugging of the system. The main modules shall include User Management and Role-Based Access Control, Customer Information Management, Order Management and Payment Recording, Seminar Management, Certified Breeder Registry, Inquiry Management, Historical Data Management, and Analytics and Reporting.

Modifications to one module shall not affect the operation of the other modules. The system shall employ consistent coding techniques, organized structures, and documented techniques to support the future maintenance and develop the system.

The Inquiry Management module shall be restricted to capturing and assigning inquiry tickets and monitoring their status. Responses and resolutions to inquiries shall be provided via the organization’s official email.

Changes to the system source code shall be implemented in a version-controlled environment using Git to support a collaborative development environment and tracking of changes. All system updates shall be tested prior to implementation to ensure the functionality of other system features.

Portability and Compatibility

The system must be operable on mainstream web browsers on desktop, laptop, tablet, and mobile phone hardware without the need for platform-specific software to be installed. A responsive design and a mobile-first approach will ensure the system delivers the same functionality and user experience across a variety of devices and operating systems.

The user interface must be designed for use on low and mid-range mobile phones that are most common among poultry farmers in the Philippines, while still accommodating larger devices. The system will be compatible with more recent versions of Google Chrome, Microsoft Edge, Mozilla Firefox and Safari.

The system will ensure that the user experience for presentation, navigation, and the system's core functionality is consistent across all devices and all supported browsers. The system's web design will ensure that the system is deployed and accessed with no need to design and build tailored versions for specific operating systems.

c. User Requirements

Administrators and Staff (Owner/Executive, IT & Administrative Staff)

1. Manage and maintain centralized customer and farm records.
2. Process and monitor Parent Stocks, F1 products, and veterinary and other farm-related orders.
3. Review proof of payment and payment records and update payment and order statuses.
4. Manage seminar modules and enrollment, track learning progress, quiz results, seminar completion and manage certificate requests.
5. Ensure Clients/Farmers meet seminar compliance before allowing them to order Parent Stocks.
6. Manage eligibility of breeders, certification status, certificate validity, and expiration dates using the Certified Breeder Registry.
7. Receive, classify, and assign inquiry tickets, as well as search, filter, and monitor tickets and update their statuses.
8. Provide actual responses to inquiries and resolutions using the official email of the organization.
9. Create, update, and manage the Frequently Asked Questions Knowledge Base.
10. Search and reference customer records, orders, payments, seminars, inquiries, and breeder certification records.
11. Access operational dashboards to create or download reports for customers, orders, payments, seminars, breeders, and inquiry tickets.
12. Manage user accounts and assign system roles and access rights based on their job functions.
13. View system and technical activity records and logs based on their roles and access rights.
14. Manage system settings and user access as part of system administration based on their authorizations.

15. View and analyze reports to assist with the monitoring and control of business operations.

Farmers and Clients (Client Role)

1. Create an account and log in to the system.
2. View and edit their customer and farm profiles with the information allowed.
3. Access and take the required seminar modules.
4. Oversee their enrollment in seminars and their status in terms of progress, assessments, and module completion.
5. Request a seminar certificate of attendance after completing the required seminar modules. This is subject to the verification and approval of the Administrative Staff.
6. Submit orders for Parent Stocks after the required seminar prerequisites have been met.
7. Submit orders for F1 and veterinary products.
8. Upload proof of payment for Administrative validation.
9. Oversee the status of payment verification and orders.
10. View transaction details, orders, and payment history.
11. View their status in terms of breeder eligibility, certification, certificate validity, and renewal schedule.
12. Submit inquiry tickets and track the status of inquiries submitted through DACS.
13. Receive responses and resolutions for inquiries through the registered email of their account.
14. For common inquiries and for more information regarding the organization and the system, please refer to the FAQs Knowledge Base.
15. Access seminar, order, payment, inquiry ticket, and breeder eligibility and certification records through one account.

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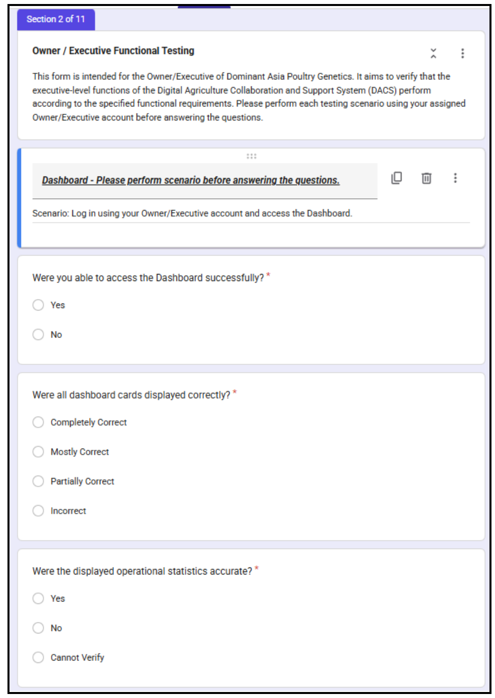
APPENDICES

This section includes the supportive documents and materials referred to during the course of the study. These provide further evidence regarding the creation, execution, and assessment of the proposed Digital Agriculture Collaboration and Support System (DACS). Provided in this section are the Letter of Endorsement, Non-Disclosure Agreement (NDA), documents pertaining to stakeholder interviews, consultations, survey questionnaires, system outputs and other materials pertinent to the study. These documents are provided to enhance the legitimacy of the research.


Section 2 of 11

Owner / Executive Functional Testing

This form is intended for the Owner/Executive of Dominant Asia Poultry Genetics. It aims to verify that the executive-level functions of the Digital Agriculture Collaboration and Support System (DACS) perform according to the specified functional requirements. Please perform each testing scenario using your assigned Owner/Executive account before answering the questions.

Dashboard - Please perform scenario before answering the questions.

Scenario: Log in using your Owner/Executive account and access the Dashboard.

Were you able to access the Dashboard successfully? *

- Yes
- No

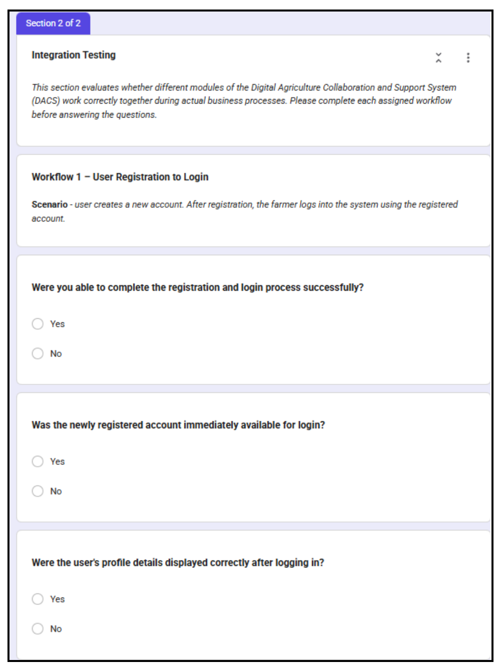
Were all dashboard cards displayed correctly? *

- Completely Correct
- Mostly Correct
- Partially Correct
- Incorrect

Were the displayed operational statistics accurate? *

- Yes
- No
- Cannot Verify

Figure A.1. Sample Google Forms Used for the Functional Testing


Section 2 of 2

Integration Testing

This section evaluates whether different modules of the Digital Agriculture Collaboration and Support System (DACS) work correctly together during actual business processes. Please complete each assigned workflow before answering the questions.

Workflow 1 – User Registration to Login

Scenario - user creates a new account. After registration, the farmer logs into the system using the registered account.

Were you able to complete the registration and login process successfully?

- Yes
- No

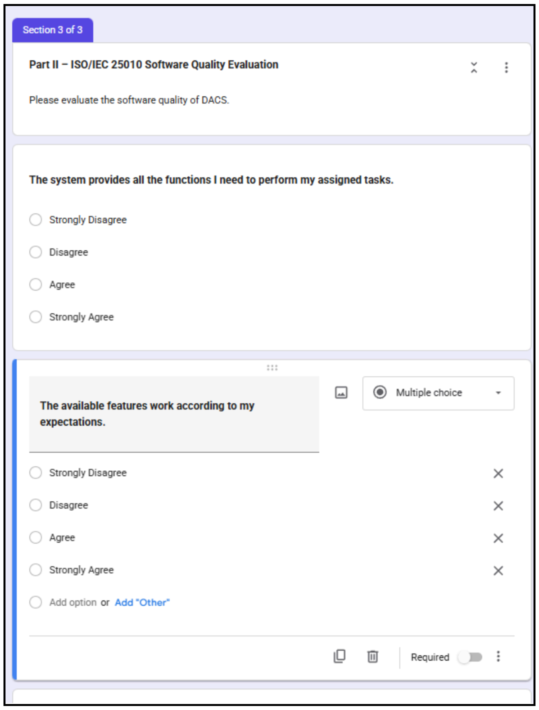
Was the newly registered account immediately available for login?

- Yes
- No

Were the user's profile details displayed correctly after logging in?

- Yes
- No

Figure A.2. Sample Google Forms Used for the Integrational Testing


Section 3 of 3

Part II – ISO/IEC 25010 Software Quality Evaluation

Please evaluate the software quality of DACS.

The system provides all the functions I need to perform my assigned tasks.

- Strongly Disagree
- Disagree
- Agree
- Strongly Agree

The available features work according to my expectations.

Multiple choice

- Strongly Disagree
- Disagree
- Agree
- Strongly Agree
- Add option or Add "Other"

Required

Figure A.3. Sample Google Forms Used for the User Evaluation Questionnaire (User Acceptance Testing and ISO/IEC 25010 Software Quality Evaluation)


LETTER OF ENDORSEMENT

Date: June 13, 2026

To Whom It May Concern:

Greetings!

This letter formally endorses the capstone project of **Adrian James M. Calalang, Ciara Frances P. Comendador, Mhikaela I. Galo, Amanda Grace R. Mangubat** from the Bachelor of Science in Information Systems (BSIS) program of CIIT College of Arts and Technology, Inc.:

The mentioned students are currently enrolled in the Capstone Project course under the supervision of **Ms. Michelle Decamora**. As part of their academic requirements, they are conducting a study aimed at designing and developing an information system that addresses real-world organizational challenges through data management, process improvement, and information technology solutions.

In connection with their study, the students may request meetings, interviews, observations, and access to relevant non-confidential information necessary for the analysis, design, development, testing, and evaluation of their proposed system. The cooperation of your organization will greatly contribute to the successful completion of their academic undertaking.

Please be assured that all information gathered shall be used strictly for educational and research purposes. Any confidential information disclosed during the course of the study shall be treated with utmost confidentiality and may be covered by a Non-Disclosure Agreement (NDA), should your organization require one.

We respectfully request your assistance and support in accommodating the students' reasonable requests related to their capstone project. Your participation will provide valuable industry insights and practical experience that will enhance the quality and relevance of their research.

Should you require any clarification regarding this endorsement or the students' project, please feel free to contact the undersigned.

Thank you for your support and cooperation.

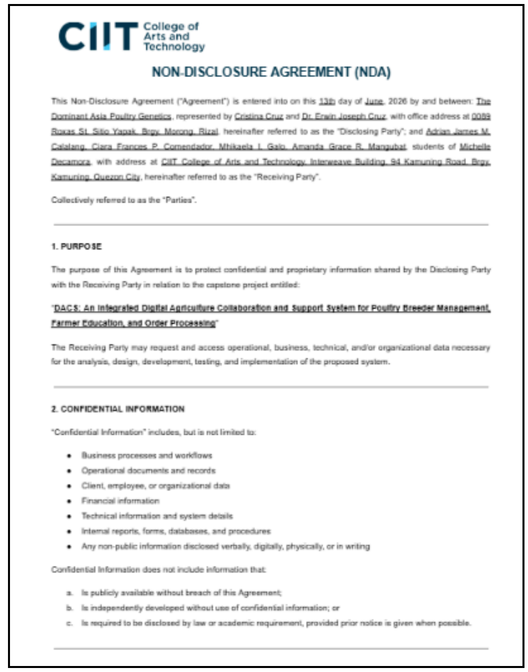
Respectfully yours,

MICHELLE DECAMORA
Capstone Adviser

Noted:

RIZALDY D. RAPSING
Program Chair, Information Systems Program

Figure A.4. CIIT Letter of Endorsement for the Proposed Capstone Project


NON-DISCLOSURE AGREEMENT (NDA)

This Non-Disclosure Agreement ("Agreement") is entered into on this 13th day of June, 2026 by and between: The Dominant Asia Poultry Genetics, represented by Cristina Cruz and Dr. Erwin Joseph Cruz, with office address at 0089 Roxas St. Sitio Yapak, Brgy. Morong, Rizal, hereinafter referred to as the "Disclosing Party"; and Adrian James M. Calalang, Ciara Frances P. Comendador, Mhikaela I. Galo, Amanda Grace R. Mangubat, students of Michelle Decamora, with address at CIIT College of Arts and Technology, Interweave Building, 94 Kamuning Road, Brgy. Kamuning, Quezon City, hereinafter referred to as the "Receiving Party".

Collectively referred to as the "Parties".

1. PURPOSE

The purpose of this Agreement is to protect confidential and proprietary information shared by the Disclosing Party with the Receiving Party in relation to the capstone project entitled:

"**DACS: An Integrated Digital Agriculture Collaboration and Support System for Poultry Breeder Management, Farmer Education, and Order Processing**"

The Receiving Party may request and access operational, business, technical, and/or organizational data necessary for the analysis, design, development, testing, and implementation of the proposed system.

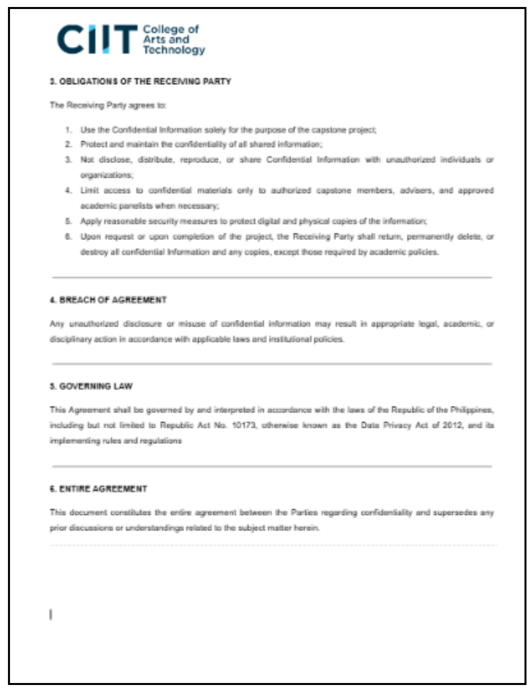
2. CONFIDENTIAL INFORMATION

"Confidential Information" includes, but is not limited to:

- Business processes and workflows
- Operational documents and records
- Client, employee, or organizational data
- Financial information
- Technical information and system details
- Internal reports, forms, databases, and procedures
- Any non-public information disclosed verbally, digitally, physically, or in writing

Confidential Information does not include information that:

a. Is publicly available without breach of this Agreement;
b. Is independently developed without use of confidential information; or
c. Is required to be disclosed by law or academic requirement, provided prior notice is given when possible.


3. OBLIGATIONS OF THE RECEIVING PARTY

The Receiving Party agrees to:

1. Use the Confidential Information solely for the purpose of the capstone project;
2. Protect and maintain the confidentiality of all shared information;
3. Not disclose, distribute, reproduce, or share Confidential Information with unauthorized individuals or organizations;
4. Limit access to confidential materials only to authorized capstone members, advisers, and approved academic panelists when necessary;
5. Apply reasonable security measures to protect digital and physical copies of the information;
6. Upon request or upon completion of the project, the Receiving Party shall return, permanently delete, or destroy all confidential Information and any copies, except those required by academic policies.

4. BREACH OF AGREEMENT

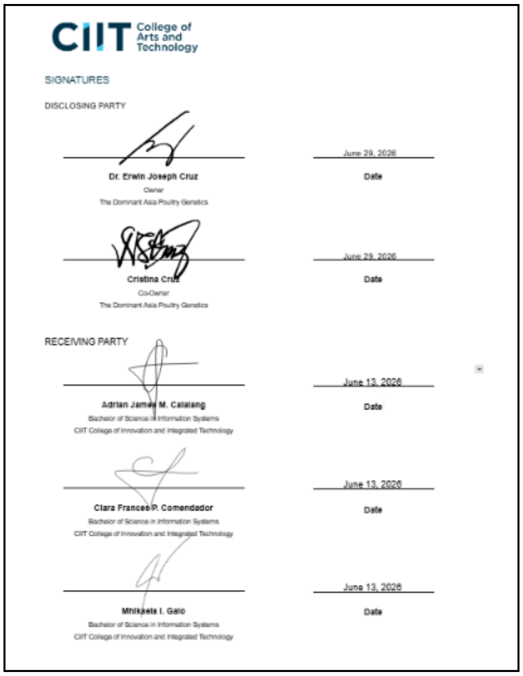
Any unauthorized disclosure or misuse of confidential information may result in appropriate legal, academic, or disciplinary action in accordance with applicable laws and institutional policies.

5. GOVERNING LAW

This Agreement shall be governed by and interpreted in accordance with the laws of the Republic of the Philippines, including but not limited to Republic Act No. 10173, otherwise known as the Data Privacy Act of 2012, and its implementing rules and regulations

6. ENTIRE AGREEMENT

This document constitutes the entire agreement between the Parties regarding confidentiality and supersedes any prior discussions or understandings related to the subject matter herein.



CIIT College of Arts and Technology

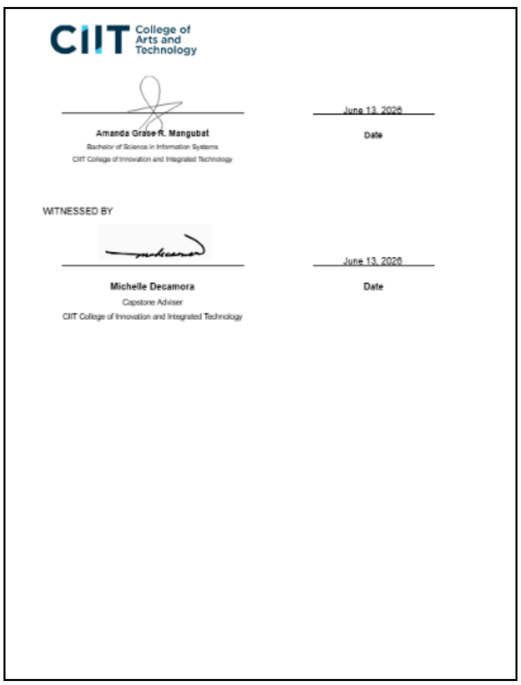
SIGNATURES

DISCLOSING PARTY

Dr. Erwin Joseph Cruz Owner The Dominant Asia Poultry Genetics	June 29, 2026 **Date**
Cristina Cruz Co-Owner The Dominant Asia Poultry Genetics	June 29, 2026 **Date**

RECEIVING PARTY

Adrian James M. Calalang Bachelor of Science in Information Systems CIIT College of Innovation and Integrated Technology	June 13, 2026 **Date**
Clara Frances P. Comendador Bachelor of Science in Information Systems CIIT College of Innovation and Integrated Technology	June 13, 2026 **Date**
Mhikaela I. Galo Bachelor of Science in Information Systems CIIT College of Innovation and Integrated Technology	June 13, 2026 **Date**



CIIT College of Arts and Technology

Amanda Grace R. Mangubat Bachelor of Science in Information Systems CIIT College of Innovation and Integrated Technology	June 13, 2026 **Date**

WITNESSED BY

Michelle Decamora Capstone Adviser CIIT College of Innovation and Integrated Technology	June 13, 2026 **Date**

Figure A.5 - A.8. Signed Non-Disclosure Agreement (NDA) Between the Proponents and Dominant Asia Poultry Genetics